Abstraction Mechanisms for Virtual Avatars

Master's Thesis

submitted by

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Abstract

Virtual reality applications increasingly rely on avatar representations to facilitate social presence and user embodiment. However, photorealistic avatar rendering can trigger the uncanny valley effect, where near-realistic human representations evoke discomfort and reduced trust rather than engagement. Non-photorealistic rendering (NPR) offers a principled alternative, deliberately abstracting avatar appearance while preserving expressiveness and identity.

This thesis presents the design, implementation, and evaluation of a real-time NPR stylisation system for virtual avatars. Eight techniques were developed iteratively: geometry-based inverted hull outlines, screen-space normal edge detection, Sobel edge detection with adaptive skin-to-clothing thresholding, Gaussian pre-filtered Sobel, hierarchical multi-cue edge detection combining depth, normal, and colour discontinuities, isotropic Kuwahara painterly filtering, and two composite techniques combining painterly abstraction with edge rendering. The techniques were implemented and compared across three avatar platforms — a Mixamo humanoid model, the Meta Avatars SDK, and Avaturn — to assess their visual behaviour and robustness under different mesh topologies and rendering pipelines. Avaturn was selected as the primary platform for the user study due to its facial rigging capabilities and higher photorealistic fidelity.

The user study investigates how visual abstraction affects the perceived trustworthiness of an avatar delivering advice. Participants watched short video scenarios in which a speaker — presented either as a real person, a photorealistic Avaturn avatar, or an NPR-stylised avatar — recommended one of two options in a practical decision situation. After each scenario, participants rated how trustworthy they found the speaker's recommendation. The results provide empirical insight into the relationship between avatar abstraction level and social trust, offering practical guidance for avatar design in applications where perceived credibility matters.

Keywords: non-photorealistic rendering, virtual reality, avatar abstraction, trust perception, edge detection, Avaturn, uncanny valley, Kuwahara filter.

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Abbreviations

ANOVA	Analysis of Variance
API	Application Programming Interface
AR	Augmented Reality
CG	Computer-Generated
ESF	Extensible Shader Framework
FPS	Frames Per Second
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HLSL	High-Level Shading Language
HMD	Head-Mounted Display
HSV	Hue, Saturation, Value (colour space)
LOD	Level of Detail
NPR	Non-Photorealistic Rendering
PBR	Physically-Based Rendering
RGB	Red, Green, Blue
SDK	Software Development Kit
SSS	Subsurface Scattering
UI	User Interface
URP	Universal Render Pipeline
UV	Texture Coordinate Axes (U and V)
VR	Virtual Reality

1 Introduction

1.1 Motivation

Communication through avatars has become a defining feature of virtual reality, extending well beyond its origins as a navigational convenience to serve as the primary medium through which users express identity, signal intent, and form social relationships in shared digital environments. The perceptual consequences of this shift are well documented. Nowak and Biocca [1] demonstrated that avatar appearance exerts a measurable influence on social presence and interpersonal trust, and Weidner and Broll [2] extended this finding to video mediated interaction, showing that the visual design of an avatar functions as a communicative act in its own right, shaping how users are perceived and how they engage with one another.

Recent advances in generative AI have made photorealistic avatar creation more accessible than ever. Platforms such as Avaturn can now produce a fully rigged, expressively animated CG avatar from a single uploaded photograph, reconstructing facial geometry and surface appearance with a degree of fidelity that was difficult to achieve outside of dedicated production pipelines only a few years ago. The implicit promise of this technology is that a digital representation close enough to the real person will be received as one. Yet this is precisely where a familiar perceptual problem arises. Mori termed this effect the Uncanny Valley, proposing that as a human representation approaches but falls just short of true photorealistic fidelity it tends to evoke unease rather than familiarity [3]. An avatar that comes close to realism without quite reaching it can therefore undermine the very trust it was designed to support, a claim corroborated empirically for computer generated characters by McDonnell et al. [4] and in the specific context of avatar stylisation and trust by Canales et al. [5].

Nonphotorealistic Rendering (NPR) offers a different path through this problem. Rather than simulating accurate lighting and surface detail, NPR techniques deliberately stylise the rendered output to achieve artistic or communicative goals. Gooch et al. [6] established a formal model for this approach, replacing physically accurate shading with stylised cues that communicate form without simulating light. Cel shading, edge line drawing, and painterly filtering can each abstract an avatar's appearance in ways that sidestep the Uncanny Valley, while preserving enough expressiveness and readable identity that the character remains engaging. The open question is not whether this looks appealing: it is whether a deliberately abstracted avatar can still be trusted by the people interacting with it, and whether reducing realism introduces its own perceptual costs. That question has received surprisingly little empirical attention in the context of modern production avatar systems managed by closed SDKs.

1.2 Problem Statement

Applying NPR stylisation to photorealistic avatar systems is not straightforward. Production avatar platforms manage rendering through proprietary shader architectures, material systems, and mesh topologies designed for physical fidelity rather than artistic abstraction, and each platform imposes a different set of constraints on what effects can be achieved. The Meta Avatars SDK represents the most restrictive case: it computes subsurface scattering, eye glints, and hair rendering through a closed Extensible Shader Framework (ESF) that manages the full lighting pipeline internally, exposing only a single fragment level hook, evaluated after all lighting computation, to third-party developers [7]. Custom NPR effects must therefore be integrated entirely within this hook, without access to the core lighting stages, without additional render passes, and without disrupting the SDK's level of detail management or material assignment system. Other production platforms provide full access to the underlying render pipeline but introduce different obstacles; textures derived from single photographs encode the photometric conditions of the capture environment, producing luminance gradients across smooth skin surfaces that undermine edge detection based on image gradients in ways that do not arise with conventionally painted assets. Prior NPR research has not had to address constraints of this kind; the literature, represented by surveys such as Isenberg et al. [8] and controlled rendering studies such as McDonnell et al. [4], assumes full pipeline access and custom assets under complete authorial control. How NPR techniques behave across production avatar platforms with substantially different integration constraints, and which technique choices remain viable under the most restrictive of them, is not established.

Beyond this technical challenge, there is an open empirical question. Photorealistic avatar generation has become increasingly accessible, yet the Uncanny Valley problem suggests that realism alone does not guarantee a positive perceptual reception. Whether NPR abstraction can improve how an avatar is perceived in terms of trustworthiness, and how much stylisation is viable before credibility is affected, are questions that remain unanswered for production avatar systems of this kind. They are consequential for any domain in which avatar credibility is at stake: social VR, remote consultation, training simulation, and education.

This thesis addresses both challenges by developing and integrating NPR techniques across three avatar platforms with different rendering constraints, and by evaluating the perceptual consequences of that abstraction in a controlled user study of trust.

1.3 Research Questions

The following three research questions guide this work:

RQ1: Can NPR stylisation techniques be implemented and applied across avatar platforms with different rendering architectures, specifically Mixamo, the Meta Avatars SDK, and Avaturn, and what technical constraints does each platform impose on the achievable set of effects?

RQ2: Across the implemented techniques, which produces the most effective visual abstraction when applied across avatar types, balancing visual quality, expressive legibility, and rendering performance?

RQ3: How does the level of visual abstraction affect the perceived trustworthiness of an avatar delivering a recommendation?

1.4 Contributions

This thesis makes the following contributions:

- **NPR shader system across three avatar platforms.** An NPR stylisation system is developed and applied across three avatar platforms: a Mixamo humanoid model used for initial technique development; the Meta Avatars SDK, integrated via the fragment hook evaluated after the lighting stage (`AppSpecificPostManipulation`); and Avaturn, which serves as the photorealistic baseline in the user study. Implementation across the three platforms reveals the constraints specific to each rendering pipeline and demonstrates how NPR techniques can be adapted to different mesh topologies and material systems.
- **Iteratively developed stylisation techniques.** A series of stylisation techniques are designed, implemented, and compared through an iterative development process: toon shading with a quantised diffuse ramp [9]; X-Toon extended toon shading with a 2D ramp lookup indexed by depth [10]; screen space normal derivative edge detection; Sobel edge detection [11] with adaptive thresholding to distinguish skin from clothing; Gaussian prefiltered Sobel [12]; hierarchical edge detection combining depth, normal, and colour discontinuities; isotropic Kuwahara painterly filtering [13]; Kuwahara combined with Sobel edge lines; and a composite technique combining Kuwahara filtering, Gaussian Sobel, and hierarchical edge detection. Each technique addresses limitations observed in its predecessor, and all are selectable at runtime without recompilation. A final technique is selected for the user study based on visual quality assessment across all three avatar platforms.
- **In-VR parameter tuning interface.** A fully procedural WorldSpace UI panel is implemented in Unity, enabling live exploration of all shader parameters from inside the headset using controller raycasting. This tool was central to the iterative development process and can be reused for future shader experimentation.
- **User study on avatar abstraction and trust.** A user study with a within subjects design is conducted using prerecorded video scenarios in which a speaker appears as either a real person, a photorealistic Avaturn avatar, or an avatar rendered with NPR stylisation, each recommending one of two practical options. The video format ensures that all three conditions are observed under identical scene and performance contexts, enabling a clean comparison across representation types. The study provides empirical evidence on how visual abstraction affects the perceived trustworthiness of an avatar's advice.

1.5 Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2 surveys prior work on NPR for 3D characters, avatar rendering and realism in VR, and trust and social presence in virtual environments, positioning this thesis within the existing research landscape.

Chapter 3 introduces the technical and theoretical foundations required for this work, covering edge detection mathematics, the Kuwahara filter, the Meta Avatars SDK rendering pipeline, and relevant concepts of trust and abstraction from human computer interaction research.

Chapter 4 details the research design, elaborating on the three research questions above, and presents the overall system architecture, the iterative development strategy, and the user study design including stimuli, procedure, and measurement instruments.

Chapter 5 describes the full implementation of the NPR shader system, covering all stylisation techniques in detail, the runtime parameter tuning interface, and the rationale for the final technique selection.

Chapter 6 reports the technical evaluation of each stylisation technique and presents the results of the user study, followed by a discussion of findings and their implications.

Chapter 7 summarises the contributions of this thesis, reflects on its limitations, and identifies directions for future work.

e

2 Related Work

This chapter surveys the literature most relevant to the central research questions of this thesis. The investigation of Non-Photorealistic Rendering (NPR) as an abstraction mechanism for production avatar systems sits at the intersection of three established areas: the perceptual and affective consequences of avatar realism and the Uncanny Valley phenomenon; trust, credibility, and social presence in virtual agent interactions; and the technical design space of NPR rendering for animated human figures. The chapter concludes by identifying the specific gap that this thesis addresses.

2.1 The Uncanny Valley and Avatar Realism

The Uncanny Valley hypothesis was first articulated by Mori, who proposed that human affinity toward an artificial figure increases with its degree of human likeness up to a critical threshold, at which point subtle imperfections cause a sharp drop in comfort [3]. The drop persists until the figure achieves complete human fidelity, after which affinity recovers. Originally framed in the context of physical robots, the hypothesis has since been applied extensively to computer-generated (CG) characters and virtual avatars.

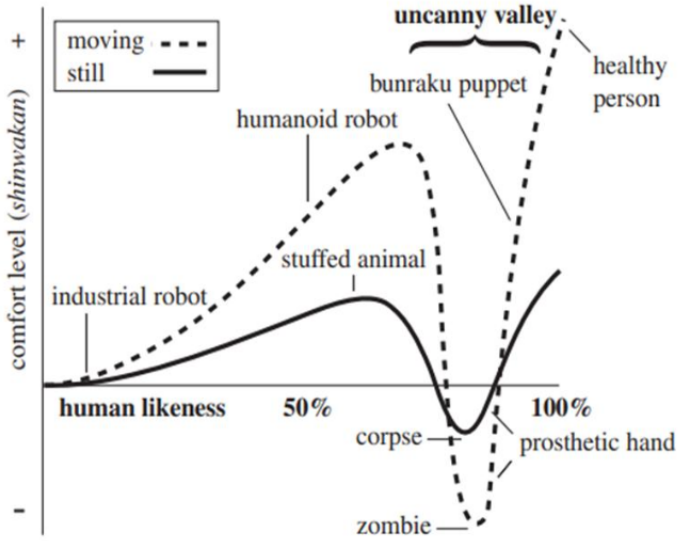


Figure 2.1: The Uncanny Valley as originally illustrated by Mori [3]. The vertical axis represents comfort level (*shinwakan*); the horizontal axis represents degree of human likeness. The solid curve describes affective response to still figures; the dashed curve describes figures in motion.

As illustrated in Figure 2.1, Mori plots two curves rather than one. Both follow the same general shape, rising as human likeness increases before dropping sharply in the high-fidelity region and recovering only at full human resemblance. The critical observation is that the curve for moving figures descends far deeper into negative comfort than the curve for still ones. The reference points Mori places inside the valley, a corpse, a zombie, and a

prosthetic hand, share a common characteristic: each resembles a human closely across most perceptual dimensions but fails on some dimension that perception detects without effort. For animated digital avatars this implication is direct. A figure at roughly sixty to seventy percent human likeness, the region occupied by a stuffed animal or a humanoid robot, sits on the safe rising slope before the drop. A figure at eighty to ninety-five percent, realistic enough to raise expectations of full human fidelity yet not realistic enough to satisfy them, occupies the most problematic region of the curve, and motion makes that positioning considerably worse.

Furthermore, MacDorman and Ishiguro advanced the Uncanny Valley from a qualitative hypothesis into an empirically tractable research programme, arguing that androids offer a uniquely controllable stimulus for cognitive and social science research precisely because they can be positioned at any point along the human likeness continuum with experimental control [14]. As part of this methodological contribution, they proposed semantic differential instruments to quantify eeriness and strangeness as measurable affective dimensions, establishing a foundation that transformed the valley from a speculative curve into a quantifiable perceptual response. They further identified the mechanism underlying the valley effect as categorical perception: when an agent occupies an ambiguous position between the human and nonhuman categories, the cognitive mismatch this ambiguity produces triggers a negative affective response. Because androids can be systematically positioned at any point along the human likeness continuum, this categorical account transfers directly to digital avatar research, where any near human representation, including a photorealistic avatar, risks eliciting the same response if its rendering does not resolve the human-nonhuman categorical boundary.

The perceptual consequences of render style choice for virtual humans were investigated systematically by McDonnell et al., who exposed participants to video stimuli of the same animated character rendered across six distinct styles spanning the range from photorealistic to fully stylised [4]. Photorealistic rendering was associated with significantly higher eeriness ratings and lower appeal than stylised conditions, while NPR styles consistently achieved more positive affective responses across both dimensions. McDonnell et al. thereby established that stylised rendering is not merely a technical shortcut but a principled perceptual strategy for sidestepping the Uncanny Valley, a finding that constitutes the primary empirical motivation for the NPR approach pursued in this thesis [4].

Modern avatar systems sit squarely in the region of the uncanny valley curve that is most problematic [3]. They achieve surface realism sufficient to trigger high perceptual expectations, but cannot fully meet those expectations in motion timing, facial microexpressions, or gaze behaviour. The practical consequence is that a photorealistic avatar may undermine the trustworthiness it was designed to convey. Whether state of the art production avatars have crossed the valley was investigated by Seymour et al. in a field study at SIGGRAPH, which compared a near photorealistic digital human to a caricature avatar [15]. Participants rated the photorealistic avatar as more trustworthy and expressed greater affinity toward it; however, the caricature condition did not trigger Uncanny Valley discomfort, occupying instead a stable, neutral region of the affinity curve. This dissociation suggests that clearly stylised avatars avoid eeriness without sacrificing social

legibility, and that trust advantages associated with photorealism are driven by factors beyond the mere avoidance of the valley.

2.2 Trust and Credibility in Virtual Agents

Trust in human-agent interaction is a multidimensional construct. The foundational theoretical model for this study is the integrative model proposed by Mayer et al., who defined trustworthiness as comprising three independently evaluable components: ability, the trustee's domain-specific competence; benevolence, the degree to which the trustee is genuinely oriented toward the trustor's interests; and integrity, the trustee's adherence to acceptable moral principles [16]. Mayer et al. further distinguished trustworthiness from trust itself, defining trust as the willingness to be vulnerable to the actions of another party based on the expectation that the party will act appropriately, irrespective of the ability to monitor or control that party. McKnight and Chervany extended this conceptual structure into a typology separating trusting beliefs from trusting intentions, allowing trust measurement to span both attitudinal and behavioural dimensions [17]. This distinction is consequential for avatar research: rendering style could plausibly affect the three trustworthiness components differently. NPR abstraction could, for example, reduce perceived competence by signalling an informal or non-professional aesthetic, while leaving perceived integrity intact because the verbal content of the recommendation is unchanged. Collapsing these three components into a single composite score would mask precisely this kind of differential effect. The three-subscale structure is therefore the appropriate operationalisation for a study that investigates rendering style rather than task framing or verbal content as its independent variable.

Alipour et al. conducted a PRISMA-compliant systematic review of 53 empirical studies at the intersection of Uncanny Valley effects and trust [18]. Their novel contribution is a taxonomy of how trust has been operationalised across the literature, distinguishing three approaches: direct trust measures, where participants explicitly rate trust or trustworthiness ($n = 24$ studies); trust-related constructs such as competence, credibility, and acceptability ($n = 11$ studies); and trust-related intentions, capturing behavioural indicators such as advice following or willingness to interact ($n = 18$ studies). Among the 41 studies that empirically tested the UVE–trust relationship, 17 reported a direct negative effect, 24 found the relationship conditional on moderating factors including task type, age, and familiarity with the agent, and 10 found no significant effect. A critical observation is that 42 of the 53 included studies employed indirect encounters, meaning participants passively viewed static images or video rather than engaging in live interaction. The majority relied exclusively on subjective self-report. Alipour et al. conclude that well-designed future studies should employ dynamic video stimuli and include at least one behavioural indicator alongside self-report ratings, a prescription that directly informs the evaluation design of this thesis [18].

Nowak and Rauh provided an early empirical foundation for the relationship between avatar appearance and credibility in online interaction [19]. In a study where 255 participants evaluated a series of avatars, perceived anthropomorphism and avatar gender expression both predicted credibility judgments, with more anthropomorphic avatars

rated as more credible. The study established that avatar design choices constitute meaningful social signals rather than cosmetic variation, and that participants apply social heuristics to avatar appearance in ways that influence their willingness to engage and trust. Nowak and Biocca extended this framework by demonstrating that anthropomorphism and perceived agency contribute independently to social presence and copresence, and that very high levels of visual anthropomorphism do not consistently produce higher social presence [1]. The implication is that avatar realism operates under diminishing perceptual returns: beyond a certain threshold of human-likeness, additional anthropomorphic cues do not translate into proportionally stronger social engagement, and may in some configurations reduce it. This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity [3].

The direct effect of avatar stylisation on multi-component trust was investigated by Canales et al., who presented participants with virtual doctors rendered at three stylisation levels: realistic, semi-realistic, and caricatured [5]. Trust was measured across competence, benevolence, and integrity dimensions before and after a short interaction, and a behavioural indicator was collected via a forced-choice selection of which doctor the participant would consult for a follow-up. Self-reported trust ratings were broadly equivalent across all three style levels; however, the semi-realistic condition attracted the highest rate of behavioural selection, suggesting that moderate stylisation can match or exceed the behavioural trust of full photorealism. The dissociation between explicit ratings and behavioural preference implies that render style influences implicit social judgment in ways not fully captured by self-report alone.

Convergent evidence was provided by Gao et al., who compared realistic and stylised virtual agents across trust, social presence, and perceived empathy [20]. Stylised agents achieved trust levels statistically comparable to realistic agents on all measured dimensions, while voice characteristics were found to have a negligible effect on trust outcomes. Gao et al. also identified perceived agent age as a significant moderator: elder-appearing agents were trusted more across both style conditions, independent of render style [20]. This result implies that social characteristics of the avatar, such as perceived demographics, can function as independent trust cues that operate above and beyond the effects of rendering style. Taken together, these results establish that NPR stylisation does not constitute a trust penalty relative to photorealism.

Weidner and Broll examined how avatar visual fidelity interacts with self-embodiment and social perception in video conferencing and VR contexts, finding that fidelity effects on social judgments are mediated by the observer's sense of identification with the avatar [2]. This has methodological implications for comparative user studies: a participant's prior familiarity with a particular avatar style can influence trust ratings independently of the stylisation itself, motivating careful stimulus counterbalancing in the evaluation design.

The Godspeed Questionnaire Series, developed by Bartneck et al., provides validated semantic differential scales for five perceptual dimensions: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety [21]. Originally developed for human-robot interaction, the series has been widely adapted for virtual avatar studies.

The likeability subscale functions as an affective trust indicator, capturing warmth and appeal as a proxy for social acceptance. The anthropomorphism subscale is particularly informative in this context because it measures the perceptual mechanism through which rendering style is hypothesised to modulate eeriness: if NPR reduces eeriness by reducing perceived human-likeness, the anthropomorphism subscale would detect that effect directly; if NPR reduces eeriness while leaving perceived human-likeness intact, that is a stronger and more theoretically significant result. Measuring anthropomorphism and eeriness independently makes this mechanism empirically testable.

The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly [14]. Ho and MacDorman subsequently refined this operationalisation into a two-subscale instrument that distinguishes perceived eeriness from perceived human-likeness, validating it across multiple stimulus types and establishing it as a standalone alternative to the Godspeed eeriness items [22]. Their principal finding is that the two dimensions are orthogonal: the reported correlation between eeriness and human-likeness in their validation data is approximately zero ($r \approx .02$). This orthogonality has a direct methodological consequence for the evaluation design of this thesis. If NPR abstraction reduces eeriness while simultaneously reducing perceived human-likeness, the two effects are entangled; it remains unclear whether eeriness decreased because the stylised avatar is less unsettling, or merely because it reads as less human. Measuring the two dimensions separately makes this question answerable: a reduction in eeriness (B_score) unaccompanied by a corresponding reduction in anthropomorphism (D_score) from C2 to C3 would indicate that stylisation reduces discomfort through a mechanism other than a simple shift in perceived humanness, a stronger and more theoretically significant result. Measuring them as a single composite would foreclose this distinction. The recommendation from Alipour et al. to include a behavioural indicator alongside self-report measures is addressed in the evaluation design of this thesis through a forced-choice preference task, which captures implicit trust judgment that explicit Likert ratings may not fully reflect [18]. All instruments described in this section are employed in the user study.

2.3 Non-Photorealistic Rendering for Virtual Humans

Non-Photorealistic Rendering encompasses a family of techniques that deliberately depart from physical light simulation, abstracting surface appearance through stylisation, edge detection, quantised shading, or painterly filtering. In the context of virtual human avatars, NPR is not a single technique but a design space with multiple technical dimensions, each producing a distinctive perceptual register with different implications for avatar appearance and social reception.

The foundational contribution to NPR lighting is the cool-to-warm shading model introduced by Gooch et al., which replaced the continuous Lambertian diffuse term with a hue shift from cool shadows to warm highlights, mimicking the conventions of technical illustration [6]. While the model targets engineering drawings rather than characters, its core insight, that shading can communicate form without physically simulating light, became a reference point for subsequent stylised rendering work.

Silhouette rendering has been approached from two geometrically distinct directions. Geometry-based methods generate outlines using the GPU rasteriser itself. The inverted hull technique described by Lake et al. expands back-facing polygons along vertex normals by a small offset, creating a slightly enlarged shell of the mesh; when front-facing geometry is rendered on top, only the protruding rim remains visible as an outline [9]. The approach requires no image-space computation, scales predictably with polygon count, and remains consistent regardless of camera distance, properties that make it well suited to animated characters in real-time applications. Isenberg et al. provide a systematic survey of silhouette algorithms and conclude that geometry-based methods are robust and distance-independent but structurally incapable of detecting internal surface detail [8]. This structural limitation is particularly consequential for avatar rendering, where facial features, clothing folds, and skin creases all constitute expressive visual cues that lie within, rather than on, the silhouette boundary.

Image-space methods address this limitation directly by operating on the rasterised output rather than on the mesh, making them sensitive to any visible discontinuity on screen regardless of whether it falls on the silhouette or in the interior of the figure. Screen-space normal derivatives, computed via hardware-accelerated `ddx`/`ddy` intrinsics available on all modern GPUs, detect abrupt changes in surface orientation without additional draw calls or intermediate render targets. This approach captures surface creases and mesh boundaries that geometry expansion cannot reach. The fundamental limitation is that edge thickness is measured in screen pixels rather than world-space units, causing edges to scale with camera distance. In an immersive VR context where the user moves relative to other participants, this produces edges that appear overly thick at close range and invisible at distance.

Texture-based edge detection targets the painted surface layer. The Sobel operator, a discrete gradient approximation applied to the diffuse texture via a 3×3 convolution kernel, identifies luminance boundaries corresponding to clothing patterns, facial contours, and painted surface details [11]. Unlike geometry-based methods, it responds to what was painted onto the surface rather than to the surface shape itself. Marr and Hildreth demonstrated that low-pass pre-filtering is a prerequisite for stable gradient computation, an observation that motivates the Gaussian pre-filtered variant explored in this work [12].

Extended toon approaches have explored richer parameter spaces. Barla et al. introduced X-Toon, which parameterises the shading tone ramp by both view angle and depth, enabling a two-dimensional lookup that encodes artistic intent unavailable in a one-dimensional quantisation step [10]. On the painterly side, the Kuwahara filter has been applied to NPR as a means of producing region-based painterly abstraction without explicit edge detection. Kyprianidis et al. extended the isotropic Kuwahara with an anisotropic formulation aligned to local image structure, producing brush-stroke effects oriented along salient contours; the anisotropic extension requires a structure tensor computed from a prior render pass, a constraint directly relevant to the integration target of this thesis [13]. Praun et al. demonstrated real-time hatching as a complementary branch of this design space, in which tonal regions are conveyed by oriented stroke textures rather than flat fills or edge lines [23].

The perceptual impact of NPR technique choices on avatar social reception has been studied in VR contexts. Zibrek et al. conducted a within-participants VR study in which the same animated character was presented in photorealistic and non-photorealistic render conditions while participants evaluated emotional recognition, social presence, and eeriness [24]. Photorealism did not significantly improve emotional recognition or social presence; in some conditions the NPR condition was rated more positively, challenging the assumption that rendering quality scales linearly with social utility. Zibrek and McDonnell extended this investigation to embodied VR, finding that photorealism significantly increased place illusion, the sense of physically inhabiting the virtual space, but had a weaker and less consistent effect on social presence [25]. This dissociation suggests that the social judgment of an avatar, including trust, is less dependent on photorealism than on animation naturalness and behavioural coherence.

Avatar representation level across three discrete conditions, abstract, semi-realistic, and realistic, was examined by Latoschik et al. in an immersive social VR context [26]. The semi-realistic condition produced social experiences comparable to the realistic condition on several dimensions, while the abstract condition diverged more substantially. This supports the view that a threshold of representational fidelity exists above which additional photorealism yields diminishing perceptual returns, a threshold that NPR avatars may already satisfy.

What connects nearly all of this body of work is a shared implicit assumption: the developer controls the rendering pipeline. Techniques are designed and validated against custom meshes, open rendering frameworks, or research engines where shader code, render passes, and material systems can be freely modified. The most directly related prior work on the avatar side, Canales et al., applied stylisation to custom assets in an open pipeline and compared coarse-grained style levels rather than specific technical approaches [5]. This assumption of full pipeline access does not hold for production avatar SDKs, where the rendering stack is managed by the SDK provider and exposes only constrained integration points.

2.4 Research Gap

The reviewed literature establishes three relevant findings. First, NPR stylisation is a principled strategy for avoiding the Uncanny Valley, with clear empirical support from McDonnell et al. and Zibrek et al. [4], [24]. Second, stylised avatars do not carry a trust penalty relative to photorealistic avatars in controlled evaluations, as demonstrated by Canales et al. and Gao et al. [5], [20]. Third, all prior work on both NPR rendering and avatar trust perception has assumed open pipeline access, using custom assets and research engines rather than production SDK systems.

Three specific gaps remain unaddressed. First, no published work has applied fine-grained NPR stylisation within a closed, compute-skinned, LOD-managed production avatar SDK, where the developer cannot modify the core lighting pipeline, add intermediate render passes, or alter material management. The NPR techniques surveyed above were all developed and validated under full rendering stack access; whether the same

techniques can be effectively integrated under production SDK constraints, and whether their perceptual properties hold in that context, is an open question. This gap is addressed by the technical contribution of this thesis (RQ1 and RQ2).

Second, while prior work compared coarse stylisation levels, the literature has not examined the technical design space within NPR under production SDK constraints. Which technique families remain viable when render-pass access and material management are controlled by an SDK, and which best preserves avatar legibility while achieving perceptual abstraction, has not been evaluated. This gap is addressed by the iterative technique development and selection described in Chapters 4 to 6 (RQ2).

Third, no prior study has measured the trust and eeriness consequences of NPR stylisation applied to a photorealistic production avatar under a controlled evaluation protocol that uses validated multi-component instruments and a behavioural measure. Canales et al. [5] is the closest prior work, but compared coarse style categories rather than a specific NPR shader pipeline, used a between-subjects design, and did not measure eeriness or anthropomorphism as separate dimensions. This gap is addressed by the user study of this thesis (RQ3), which tests five hypotheses: that NPR reduces eeriness relative to photorealism (H1); that NPR does not reduce self-reported trust (H2); that both avatar conditions produce lower trust than a real human (H3); that eeriness negatively predicts trust across conditions (H4); and that anthropomorphism mediates the condition-to-eeriness path (H5, exploratory).

This thesis addresses all three gaps. It integrates a suite of NPR techniques into the Meta Avatar SDK under real production constraints, evaluates the visual and perceptual trade-offs across technique variants, and measures the trust and eeriness implications of NPR abstraction in a within-subjects, video-based user study. The study presents three conditions: a recording of a real person serving as the trust ceiling baseline (C1), an Avaturn avatar rendered at full photorealistic quality (C2), and the same Avaturn avatar rendered with the selected NPR shader pipeline (C3). Holding the avatar platform constant between C2 and C3 isolates the perceptual effect of NPR stylisation from platform variation. The evaluation instrument combines the Godspeed likeability and anthropomorphism subscales [21], eeriness scales grounded in MacDorman and Ishiguro [14] and refined by Ho and MacDorman [22], and a multi-component trust battery structured around the ability, benevolence, and integrity dimensions of the Mayer et al. integrative trust model [16], operationalised within the trusting-beliefs typology of McKnight and Chervany [17]. A forced-choice behavioural preference task supplements the self-report battery, following the paradigm of Canales et al. [5] and the methodological prescription of Alipour et al. [18].

3 Background

This chapter introduces the technical and theoretical foundations required for the work described in subsequent chapters. It covers the concept of virtual avatars and visual representation, edge detection mathematics, the Kuwahara painterly filter, and the human factors concepts of trust and social presence relevant to the user study.

3.1 Mathematical Notation

Throughout this thesis, scalars are denoted by italic letters x, vectors by bold lowercase letters $\mathbf{x}$, and matrices by bold uppercase letters $\mathbf{X}$. The i-th element of vector $\mathbf{x}$ is written x_i. Convolution of a signal I with a kernel K is written $K * I$. In shader pseudocode, `ddx(f)` and `ddy(f)` refer to the hardware-computed screen-space partial derivatives of a value `f` with respect to the horizontal and vertical screen axes, respectively.

3.2 Edge Detection

An edge in an image is conventionally defined as a location where intensity changes rapidly. Detecting such locations involves estimating the spatial gradient of the image intensity function, then applying a threshold to identify pixels where the gradient magnitude exceeds some criterion.

3.2.1 Gradient-Based Detection

Given an image $I(x, y)$, the gradient is the vector $\nabla I = (\partial I/\partial x,\, \partial I/\partial y)$. Its magnitude

$$M = \|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2} \tag{3.1}$$

provides a scalar measure of edge strength at each pixel. In practice, the partial derivatives are approximated by discrete convolution with finite-difference kernels.

The Sobel operator [11] estimates the gradient using two 3×3 kernels:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{3.2}$$

G_x responds to horizontal intensity transitions (vertical edges); G_y responds to vertical transitions. The gradient magnitude is $M = \sqrt{G_x^2 + G_y^2}$, computed pointwise after convolving the image with each kernel separately. The ± 2 weights along the central row and column give the Sobel operator a small amount of spatial smoothing relative to the simpler Prewitt kernel, improving noise robustness marginally.

The Roberts Cross operator uses a 2×2 diagonal kernel pattern instead:

$$R_1 = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad R_2 = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}. \tag{3.3}$$

The edge magnitude is $|R_1| + |R_2|$, which approximates the gradient diagonal without a centre sample. Roberts Cross is faster and more local than Sobel but more sensitive to noise. In this thesis it is used specifically for colour edge detection within the hierarchical technique, where its two-sample footprint keeps the per-fragment sample count low.

3.2.2 Screen-Space Normal Derivatives

Modern GPU pipelines expose per-fragment screen-space derivatives of any interpolated value via the `ddx` and `ddy` intrinsics, computed using the finite difference between adjacent 2×2 pixel quads. Applying this to the world-space surface normal $\mathbf{N}$ gives

$$M_N = \sqrt{\|\texttt{ddx}(\mathbf{N})\|^2 + \|\texttt{ddy}(\mathbf{N})\|^2}, \tag{3.4}$$

which responds to abrupt changes in surface orientation — creases, mesh boundaries, and regions of high curvature. This computation requires no additional texture samples and executes at negligible cost relative to the surrounding fragment shader. The significant practical limitation is that screen-space derivatives are measured in pixel units. A fixed-world-space edge — the crease between a jacket lapel and its collar, for instance — projects to a varying number of pixels depending on camera distance, causing edge thickness to scale with the viewpoint rather than with the surface.

3.2.3 Gaussian Pre-Filtering

Differentiation amplifies high-frequency components. Marr and Hildreth [12] established that stable edge detection requires low-pass smoothing before gradient computation. Because convolution and differentiation are both linear operations, the smoothed gradient can be written as

$$\nabla(G_\sigma * I) = (\nabla G_\sigma) * I, \tag{3.5}$$

where G_σ is a Gaussian kernel with standard deviation σ. Canny's influential edge detector [27] built on this principle: smooth with a Gaussian, compute the gradient of the smoothed image, and apply non-maximum suppression to obtain thin, stable edge maps. In a single-pass fragment shader, a true two-pass separable convolution is not available. The Gaussian can instead be approximated by sampling the source texture at a fixed set of offsets with precomputed weights, accepting that the smoothed values are recomputed independently at each sample position.

3.3 The Kuwahara Filter

The Kuwahara filter [13] is a structure-preserving smoothing operator that produces piecewise-uniform regions with sharp inter-region boundaries — a visual effect that

resembles painterly brushwork. It operates by dividing the neighbourhood around each pixel into k overlapping subregions, computing the mean and variance of pixel values within each subregion, and setting the output to the mean of whichever subregion has the lowest variance. Formally, for subregions $\{Q_i\}_{i=1}^{k}$ and their respective means μ_i and variances σ_i^2:

$$F_{\text{Kuwahara}}(x, y) = \mu_j, \qquad j = \arg\min_i \sigma_i^2. \tag{3.6}$$

The effect is that pixels on uniform surfaces are assigned the mean colour of their most locally uniform neighbourhood, while pixels near strong boundaries pick the subregion that falls entirely on one side. Edges are therefore sharpened rather than blurred, even as internal regions are smoothed toward a uniform painterly fill.

The classical isotropic formulation uses four quadrant-shaped subregions of equal size, producing a rotationally symmetric filter that does not prefer any orientation. An anisotropic extension by Kyprianidis et al. [13] aligns the filter kernel with local image structure by computing a structure tensor from a prior render pass, producing brush-stroke effects oriented along salient contours. This produces superior results visually, but requires an intermediate render target that is incompatible with a single-pass fragment hook.

3.4 Virtual Avatars and Visual Representation

In virtual environments, an avatar is a digital body that represents a user within a shared space. The term covers a wide range of forms, from abstract geometric markers to fully articulated, photorealistic human figures, and the choice of representation carries consequences that extend well beyond visual preference.

Avatar representations can be placed along a continuum of anthropomorphism and visual fidelity. At one end, abstract or stylised avatars retain the structural properties of a humanoid figure without committing to the surface detail of a real person. At the other, photorealistic avatars attempt to reproduce the user's physical appearance with enough accuracy to approach the visual fidelity of a photograph. Where an avatar sits along this continuum has measurable consequences for how users are perceived, how much social presence is felt in the interaction, and how the avatar is trusted [1], [26].

Avatar appearance also shapes the behaviour of the person embodying it. Yee and Bailenson [28] demonstrated through a series of experiments that users assigned more attractive or physically taller avatars adopted behaviours consistent with those characteristics: they disclosed more personal information, negotiated more assertively, and maintained shorter interpersonal distances. This tendency, termed the Proteus effect, illustrates that visual self-representation in virtual environments is not a passive interface element but an active influence on how users act and present themselves.

Contemporary avatar platforms have made high-quality avatar creation accessible without requiring manual artistic work. Services such as Avaturn reconstruct a personalised, fully rigged avatar from a single uploaded photograph using learned three-dimensional reconstruction, producing a mesh with facial blendshape animation support that integrates

directly into standard game engine pipelines. This accessibility makes individualized photorealistic representation practical for a wide range of applications, and it sharpens the perceptual stakes: an avatar that looks like a real person will inevitably be compared to one, and any shortfall in that comparison has consequences both for immersion and for how the user is received by others.

3.5 Trust and Social Presence in Avatar Interaction

Trust in the context of human-avatar interaction is typically understood as a social judgement: the degree to which an observer is willing to rely on the entity represented by the avatar. It is related to but distinct from social presence, which refers to the subjective sense of sharing a space with another being. Both constructs are influenced by avatar appearance, but through different mechanisms. Social presence depends strongly on the perceived anthropomorphism of the avatar and the responsiveness of its motion [1]. Trust is additionally sensitive to perceived competence, predictability, and familiarity, including the absence of uncanny valley discomfort [3].

The dominant theoretical account of interpersonal trust in organisational and HCI contexts is the Ability-Benevolence-Integrity (ABI) model of Mayer et al. [16]. The model identifies three independent components. *Ability* refers to the trustee's perceived competence within a relevant domain: a person, or an avatar, may be trusted to give technically sound advice without being trusted to act in the observer's interest. *Benevolence* refers to the extent to which the trustee is perceived to care about the observer's wellbeing beyond its own motivations. *Integrity* refers to whether the trustee is perceived to adhere to acceptable principles, including honesty and avoidance of deception. The three components are theorised to be separable: a stylistic change to an avatar's appearance could, for example, reduce perceived competence by signalling an informal or non-professional aesthetic while leaving perceived integrity intact, since integrity judgements may rely more heavily on verbal content than on visual form. Measuring the three components separately rather than as a single composite is therefore necessary to detect these differential effects.

Measurement instruments for the ABI dimensions in human-computer interaction contexts were operationalised by McKnight and Chervany [17], whose trusting-beliefs typology provides item-level questionnaire language for each component. The instrument used in this thesis adapts McKnight and Chervany's items into the advisor-evaluation context described in Chapter 4.

A complementary set of measurement instruments focuses on perceptual responses to avatar appearance. The Godspeed questionnaire developed by Bartneck et al. [21] provides semantic differential items for five subscales: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety. Originally validated on robotic systems, it has since been applied extensively to avatar stimuli. The anthropomorphism and likeability subscales are relevant here: the anthropomorphism subscale measures how human-like a stimulus is perceived to be, and the likeability subscale provides an affective assessment complementary to the cognitive trust subscales.

Eeriness, which is the affective component most directly associated with uncanny valley responses, is not well-captured by the Godspeed items. Ho and MacDorman [22] developed a separate eeriness instrument consisting of six semantic differential items targeting strangeness, unsettlingness, and perceptual discomfort. Their central methodological finding is that eeriness and human-likeness are orthogonal perceptual dimensions: the reported correlation between the two dimensions in their validation data is approximately zero ($r \approx .02$). This orthogonality is relevant to the measurement design of this thesis, because it means that a reduction in eeriness from a photorealistic avatar to an NPR avatar cannot be explained by a simple shift in perceived humanness alone; if eeriness falls while anthropomorphism remains stable, some other perceptual mechanism is responsible.

A systematic review by Alipour et al. [18] examined the relationship between uncanny valley discomfort and trust across 42 empirical studies. Their review found a consistent negative association: stimuli that produce higher eeriness ratings tend to receive lower trust ratings, and this association is independent of whether trust is measured with cognitive or affective instruments. Alipour et al. also found that the association is strongest when stimuli are presented as dynamic video clips rather than static images, motivating the use of video stimuli in this thesis.

A consistent finding across this literature is that the relationship between visual fidelity and perceived trustworthiness is not linear. Near-photorealistic representations can produce lower trust scores than clearly stylised ones, consistent with the uncanny valley hypothesis. This non-linearity motivates the experimental design of this thesis, which tests whether NPR abstraction can reduce eeriness while leaving trust intact, rather than assuming that any reduction in realism will necessarily reduce trust.

4 Methodology

This chapter describes the research questions guiding this work, the three avatar platforms used as development and evaluation targets, the NPR stylisation techniques developed across those platforms, and the design of the user study that evaluates their perceptual consequences.

4.1 Research Questions

Three research questions structure the contributions of this thesis.

RQ1: Can NPR stylisation techniques be implemented and applied across avatar platforms with different rendering architectures, specifically Mixamo, the Meta Avatars SDK, and Avaturn, and what technical constraints does each platform impose on achievable effects?

This question is primarily technical and addresses two related sub-problems. First, it asks whether a viable integration path exists for each platform, given that the three platforms differ substantially in mesh topology, shader accessibility, and rendering pipeline structure. Second, it asks what constraints each integration path places on the set of NPR effects that can be achieved, which determines which techniques from the full design space are realisable on each platform.

RQ2: Which NPR technique provides the best trade-off between visual quality, expressiveness, and rendering performance when applied across avatar types?

For this question, *visual quality* refers to the perceptual clarity and coherence of the stylised output; *expressiveness* refers to the degree to which facial features and body contours remain readable under stylisation; and *rendering performance* refers to the per-frame GPU cost measured as frame time. This question drives the iterative development of the NPR techniques. Each technique operates on a different definition of a visual boundary, works in a different coordinate space, and incurs a different sample cost per fragment. Answering RQ2 requires both qualitative visual assessment of expressiveness and feature readability, and quantitative frame-time measurement, evaluated across all three avatar platforms. Where these two assessments conflict, expressiveness is given priority because the user study requires facial animation to remain legible.

RQ3: How does the level of visual abstraction affect the perceived trustworthiness of an avatar delivering a recommendation?

This is the central empirical question. The phrase *level of visual abstraction* refers here to the contrast between two discrete conditions: a photorealistic avatar with standard PBR shading and an NPR-stylised avatar rendered with the technique selected from RQ2. A continuous scale of abstraction would require many conditions; the study operationalises the question as a targeted comparison between these two points, with a real human

speaker providing the trust ceiling baseline. The formal hypotheses tested are stated in Section 4.7.1.

4.2 Avatar Platforms and Rendering Constraints

The NPR techniques developed for this thesis are evaluated across three distinct avatar platforms, each offering a specific combination of capabilities and integration constraints relevant to NPR development. Mixamo offers the most flexible environment for initial shader implementation due to its unrestricted access to URP shaders, but its exported characters lack facial rigging. In contrast, the Meta Avatars SDK provides native Unity integration and built-in body tracking, which enables the observation of NPR behaviour under realistic animation. However, it restricts shader customisation to a proprietary injection interface, making it the most constrained platform for NPR shader development. Finally, Avaturn combines unrestricted URP access with high fidelity facial reconstruction and full blendshape animation, though it requires an additional import step for its glTF assets. By leveraging this architectural diversity, the proposed NPR techniques can be systematically evaluated against structurally distinct avatar representations, providing a broader and more robust empirical foundation for the final comparative assessment.

Mixamo.

Figure 4.1: Mixamo: baseline avatar for NPR shader development.

The Mixamo platform provides an online library of rigged humanoid characters and skinned animations that can be exported to Unity as standard FBX assets [29]. Since Mixamo avatars are standard skinned meshes without proprietary shader restrictions, all NPR techniques can be implemented directly within the URP rendering pipeline. Consequently, Mixamo is the least constrained of the three platforms, making it an appropriate environment for the initial exploration and development of each proposed approach before adaptation to the more constrained platforms. However, Mixamo character exports do not include facial rigging, meaning the avatar's face remains static throughout all renders generated on this platform. As a result, Mixamo is unsuitable for the user study, although

the Mixamo character (Figure 4.1) served as the primary prototyping target during the initial development phase, where unrestricted shader access facilitated rapid prototyping and iterative refinement.

Meta Avatars SDK.

Figure 4.2: Meta avatar: baseline avatar for NPR integration within the Meta Avatars SDK.

The Meta Avatars SDK provides Meta's avatar framework for Quest devices, handling avatar rendering, face tracking, and body tracking internally [7]. Since one objective of this thesis is to evaluate NPR avatars in VR, and the Meta Quest 3 was selected as the target headset, the Meta Avatars SDK was examined for the shader customisation capabilities it provides for NPR integration (Figure 4.2). Its built-in face and body tracking eliminates the need for external motion capture systems, allowing the proposed NPR shaders to be evaluated immediately on naturally animated avatars in a VR environment. However, custom NPR shaders cannot be integrated through standard URP material slots due to the SDK's internal rendering architecture. Instead, NPR rendering logic must be injected through hook functions exposed by the SDK at defined points in the shader pipeline. This constrains both the range of achievable NPR effects and the structural requirements of the shader code. The technical details of this integration are described in Chapter 5.

Avaturn.

Figure 4.3: Avaturn: baseline avatar for the user study.

Avaturn is an avatar creation service that reconstructs a personalised rigged avatar with facial blendshape targets from three uploaded photographs (Figure 4.3) [30]. The service exports its output as `.glb` files, which implement the glTF 2.0 standard and bundle geometry, PBR materials, and skeletal rigging within a single self-contained file [31]. Since Unity does not natively support this format at runtime, an additional import package is required to load the avatar, as described in Chapter 5. Once loaded, it functions as a standard skinned mesh with PBR materials and unrestricted access to the URP rendering pipeline, enabling the proposed NPR shaders to be implemented directly. Consequently, Avaturn served as the primary platform for the final user study by combining full shader access with personalised avatars featuring high fidelity facial blendshape animation. This complemented the prototyping workflow established with Mixamo and the VR evaluation performed using the Meta Avatars SDK.

Table 4.1: Summary comparison of the three avatar platforms across dimensions relevant to NPR integration.

	Mixamo	**Meta Avatars SDK**	**Avaturn**
Shader access	Full URP, unrestricted	Hook injection only (SDK post lighting)	Full URP, unrestricted
Transparency	Explicit alpha cutout	SDK managed	Explicit alpha cutout
Facial animation	None	SDK managed face tracking	Blendshape animation
Key challenge	No facial rigging	Shader manipulation	Initial Unity import
Strength	Full shader freedom	Native Quest 3 integration	High fidelity facial quality

4.3 NPR Technique Design

NPR techniques in computer graphics encompass a wide range of abstraction levels [8]. Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability [32], [33]. Thus, selecting the appropriate degree of abstraction requires calibrating the balance between realism and visual expressiveness.

This calibration was performed through an iterative design process in which several NPR techniques were developed and evaluated. Each version addressed a specific limitation identified in the preceding version, and the final selection was validated through a formal user study. Main techniques are labelled V1 through V7. Extended variants carry a decimal suffix, as in V2.2 and V4.2, indicating a directed modification of the preceding version.

Development proceeded across three platforms in sequence by starting from the Mixamo platform, where unrestricted shader access enabled rapid prototyping and successive refinement. The techniques were then integrated into the Meta Avatars SDK, where built-in face and body tracking supported evaluation on animated avatars in a VR environment. The SDK's shader hook architecture, however, imposed implementation constraints that shaped several architectural decisions. Finally, the techniques were transferred to the Avaturn platform to verify that each produced comparable visual output across structurally different avatar pipelines. The following sections describe the motivation, design rationale, and visual characteristics of each approach across all three systems. Full implementation details are provided in Chapter 5.

4.3.1 Shared Rendering Structure

All proposed NPR techniques share a common two-pass rendering structure. The first pass generates the silhouette outline using the inverted hull method, which is applied consistently across all three platforms. The second pass computes the NPR surface effect for each technique within the forward rendering pass. Separating outline generation from surface shading allows the outline width, colour, and depth behaviour to be adjusted independently while ensuring that the outline geometry is rendered at a consistent depth regardless of the transparency mode of the surface material.

4.3.2 V1 — Inverted Hull Outline

Object silhouettes are a defining visual feature of nonphotorealistic rendering [8]. Accordingly, a robust outline generation technique was required as the foundation for all NPR techniques developed in this thesis. The inverted hull method, also known as the shell method, was selected because it produces stable geometric silhouettes without requiring screen space processing [8]. It generates an outline shell by enlarging back-facing polygons and relies on GPU face culling to produce the outline, eliminating the need for edge detection or additional rendering passes. The method operates across two dedicated stages:

1. **Outline Pass.** The mesh is rendered with front-face culling enabled, displacing each vertex outward along its world-space surface normal by a configurable distance w:

$$\mathbf{P}' = \mathbf{P} + \mathbf{N} \cdot w \tag{4.1}$$

   where $\mathbf{P}$ is the original vertex position and $\mathbf{N}$ is the world-space surface normal. Only the back-facing geometry is rasterised, forming an enlarged shell around the mesh.

2. **Main Pass.** The mesh is subsequently rendered with standard back-face culling. The front-facing surfaces overwrite the interior of the previously rendered shell, leaving only the protruding rim visible as a silhouette border with controllable thickness.

This approach is well suited to real time NPR avatar rendering because it requires no offline mesh preprocessing and executes entirely within the GPU vertex transformation stage [8]. As a result, its computational cost scales predictably across assets of varying complexity. However, the expanded shell and the original surface may produce similar depth values, which can introduce z-fighting artefacts along the silhouette boundary. To prevent this, a depth bias is applied to shift the outline geometry slightly towards the camera in clip space.

Despite its stability and low rendering cost, the inverted hull method has several limitations when applied to avatar rendering. Figure 4.4 shows the results across all three platforms. Surfaces with degenerate or coincident normals, such as flat planes, produce no visible outline because displacing vertices along identical surface normals does not generate an expanded shell. In addition, the method does not capture interior surface boundaries. Consequently, creases between body regions, clothing seams, and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh [8]. This limitation is shared by geometry-based outline methods and motivates the interior edge detection techniques introduced in V3.

Alpha Cutout for Transparency

Avatars on the Mixamo and Avaturn platforms contain alpha-masked elements such as eyelashes and hair, which are represented as flat cards. Their visible shape is defined by discarding fragments below an opacity threshold. Without alpha testing in the outline pass, the expanded hull would be generated from the full geometric extent of each card, producing solid rectangular borders rather than outlines that follow the visible hair silhouette, as shown in the left column of Figure 4.5. Alpha testing is therefore applied using the GPU's built-in fragment clip operation to discard fragments whose texture alpha value falls below a configurable cutoff threshold α_{cutoff}:

$$\text{if } \alpha_{\text{texture}} < \alpha_{\text{cutoff}} \quad \text{then discard} \tag{4.2}$$

This test executes in both the outline pass and the forward lit pass to maintain silhouette consistency. A threshold of $\alpha_{\text{cutoff}} = 0.1$ was selected empirically as the lowest value that reliably removes fully transparent regions while preserving the visible hair silhouette

(a) Mixamo: inverted hull silhouette outline.

(b) Meta SDK: inverted hull silhouette outline.

(c) Avaturn: inverted hull silhouette outline.

Figure 4.4: Inverted hull geometry outline applied to all three avatar platforms. The slight floor penetration visible at the Meta Avatar's feet in (b) reflects a tracking boundary offset during capture and is not a property of the outline technique.

without clipping antialiased edges. Unlike alpha blending, this binary approach does not require depth sorting and introduces negligible additional fragment processing cost.

Conversely, the Meta Avatar platform handles hair and eyelash rendering through its internal pipeline and does not require manual alpha testing during outline generation. The SDK processes this geometry before the outline hook is executed, so transparent fragments are already discarded by the time the outline colour is applied. The same two-pass rendering structure is therefore maintained on this platform, although the underlying outline masking approach differs from the explicit alpha cutout used on Mixamo and Avaturn. The full technical details are discussed in Section 5.3.2.

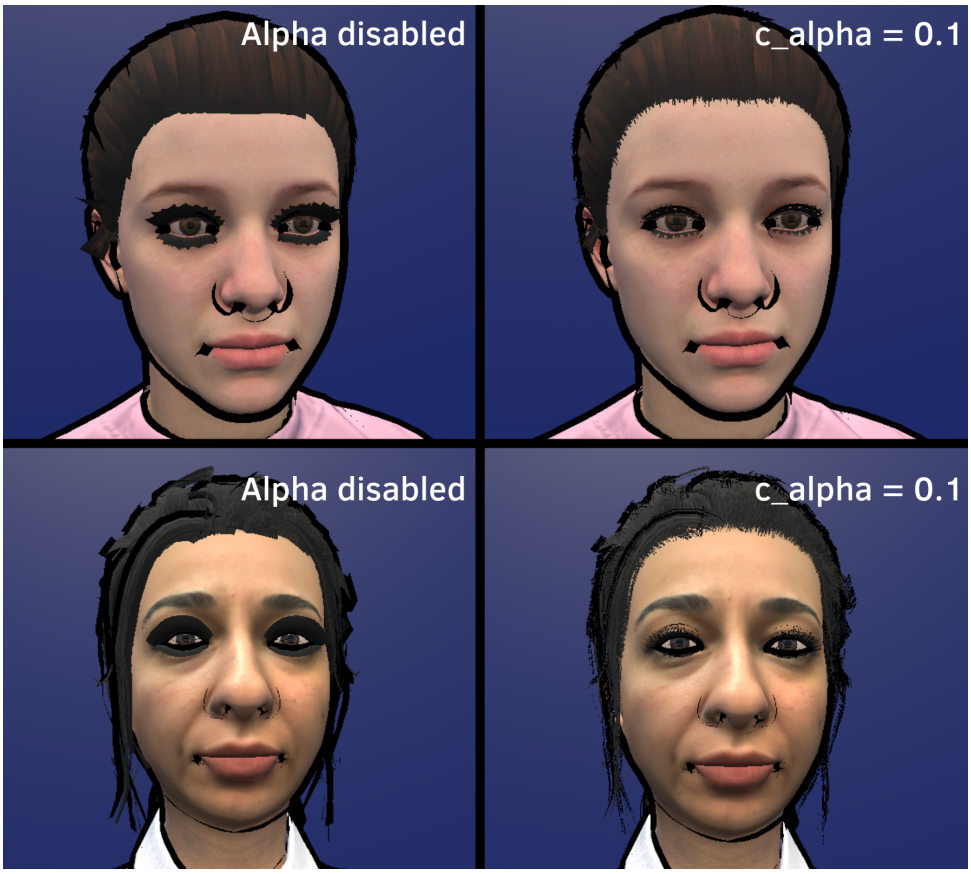


Figure 4.5: Alpha cutout on the inverted hull outline pass (Mixamo, top; Avaturn, bottom). Left: alpha testing disabled. Right: alpha cutout enabled at $\alpha_{\text{cutoff}} = 0.1$.

4.3.3 V2 — Toon Shading

Toon shading, also known as cel shading, is one of the most widely adopted techniques in NPR, where stylisation is achieved by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination [9]. Discretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand drawn animation, making the technique well suited to interactive character rendering [9]. Accordingly, toon shading was selected as the V2 baseline because it provides a controlled starting point from which each subsequent technique introduces a targeted increase in stylisation. Implementing this stylisation across the three avatar platforms, however, requires platform specific adaptations due to differences in their rendering architectures.

On Mixamo and Avaturn, the continuous illumination response is converted into discrete tonal regions using the following quantisation function:

$$t = \frac{\lfloor s \cdot n \rfloor}{n} \tag{4.3}$$

where s represents the surface lighting response and n defines the number of shading bands. This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading [9]. Four shading

bands were selected as the default configuration because visual assessment confirmed they provide an effective balance between shape legibility and graphic simplification. The resulting tonal structure consists of highlight, mid tone, shadow, and deep shadow regions.

Several refinements were introduced to improve the visual quality and temporal stability of the baseline toon shading model. Tonal region transitions were softened during animation, preserving the stepped appearance while eliminating interframe flickering at band boundaries. Additional ambient illumination was introduced to prevent heavily shadowed regions from becoming visually disconnected from the avatar. A rim highlight was also added to improve visual separation between the avatar and its background where flattened shading reduces depth cues (Figure 4.6). Finally, scene cast shadows were deliberately excluded because they introduce an additional visual variable that could interfere with evaluating the effect of tonal quantisation in isolation.

Unlike Mixamo and Avaturn, where intermediate lighting values are directly accessible, the Meta Avatars SDK exposes only the final composited appearance after its internal rendering process. Consequently, quantisation is applied to the available colour output rather than to the surface lighting response itself. Despite the different implementation, both approaches achieve the same intended stylisation by replacing smooth illumination gradients with discrete tonal regions.

Across all three platforms, V2 successfully establishes a simplified graphic appearance through tonal quantisation. However, the technique has two important limitations. First, the distribution of tonal regions remains closely coupled to the lighting response, providing limited artistic control over the tonal distribution. Second, as a purely shading-driven technique, V2 does not introduce additional edge information for interior features such as facial contours, clothing folds, or structural boundaries. These constraints motivate the subsequent development stages. V2.2 extends the tonal representation through X-Toon shading to provide greater control over the tonal distribution, while V3 introduces interior edge detection to enhance the visibility of local avatar features beyond what tonal quantisation alone can convey.

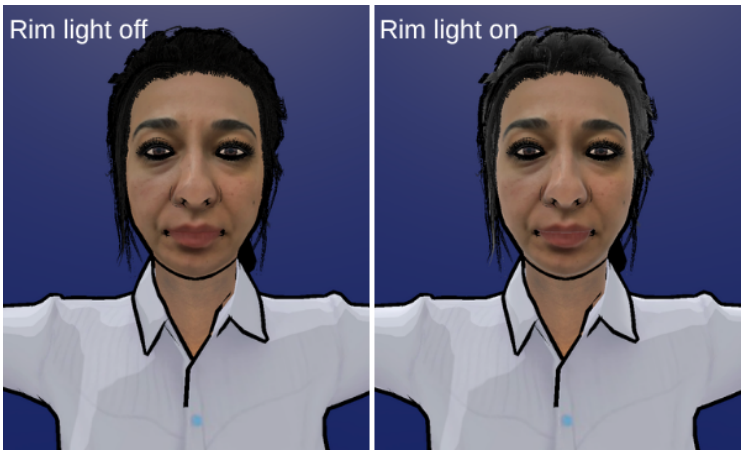


Figure 4.6: Rim highlight effect in V2 toon shading.

(a) Mixamo: four discrete shading bands with ambient and rim.

(b) Meta SDK: four discrete shading bands applied to the fully composited colour produced by the SDK renderer.

(c) Avaturn: four discrete shading bands with ambient and rim.

Figure 4.7: V2 (Toon Shading) across all three avatar platforms.

4.3.4 V2.2 — X-Toon Shading

X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level [10]. This second axis allows the surface appearance to be adjusted independently of the lighting conditions, providing finer stylisation control, and a higher abstraction level reduces local tonal contrast and softens transitions between shading regions toward a flatter, more uniform appearance. Figure 4.8 illustrates the ramp texture used in this work.

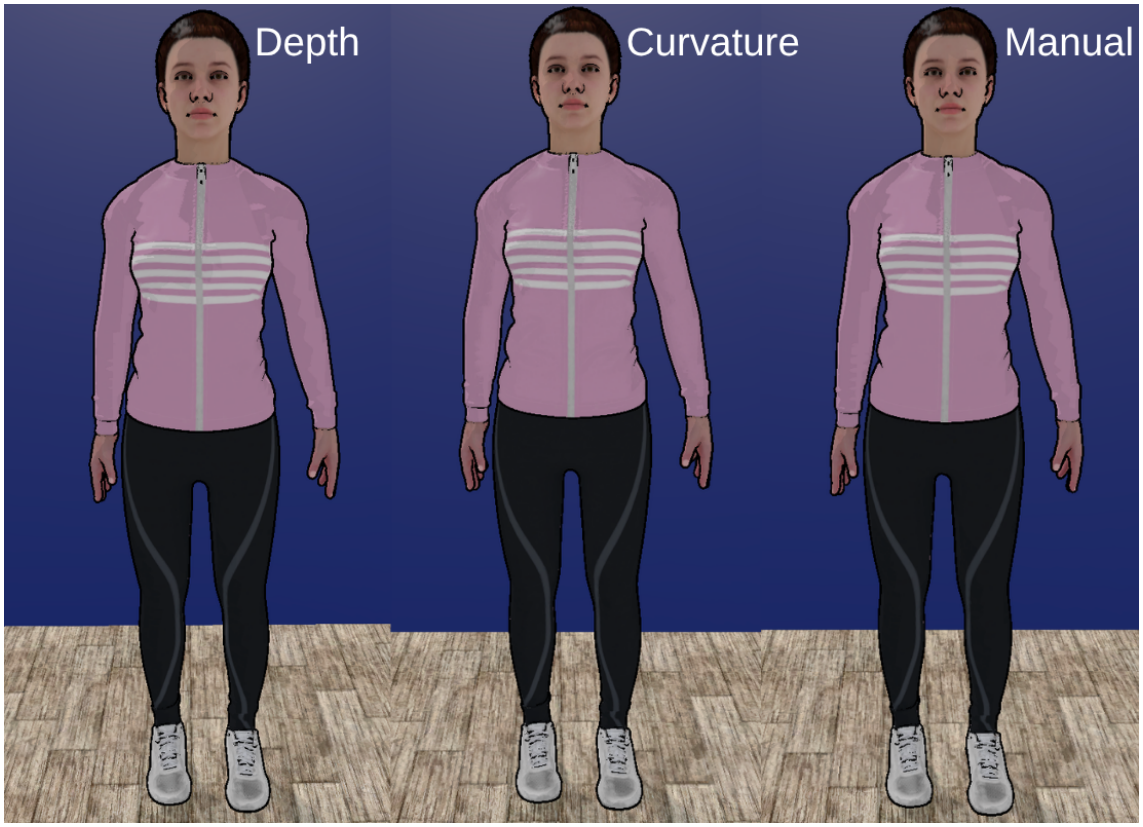
Figure 4.8: 2D ramp texture used for X-Toon shading.

In addition to tonal abstraction, Barla et al. described a shape detail extension that modifies the surface normal field by blending the original mesh normals with normals derived from a geometrically simplified shape [10]. This blending was incorporated into the Mixamo and Avaturn implementations, reducing local surface variation and producing more coherent stylised shading.

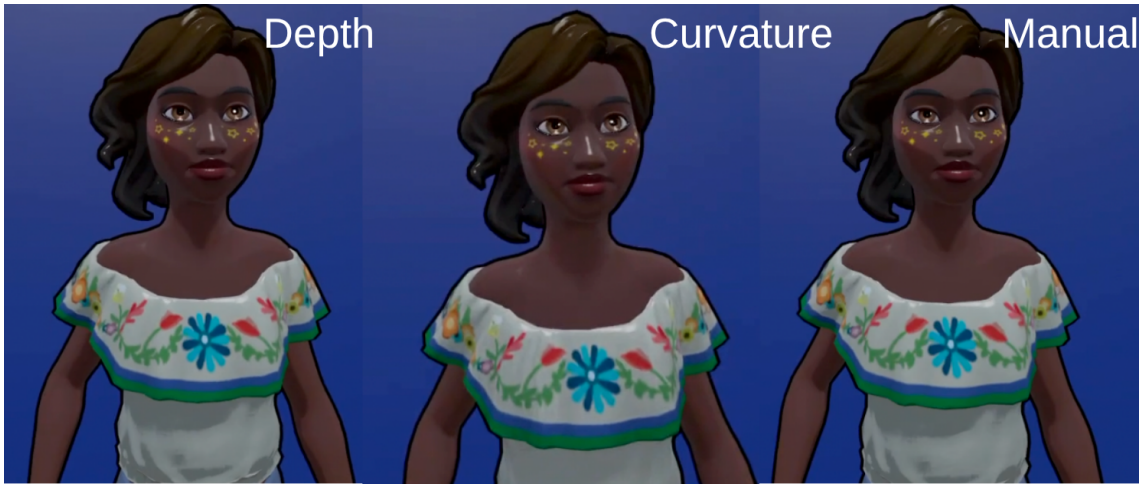
Barla et al. further described an attribute mapping mechanism that controls the abstraction

level through three driving modes: depth, curvature and manually controlled [10]. All were evaluated in this work. Visual comparison revealed only minor differences between them across the evaluated platforms, although Avaturn exhibited slightly greater variation due to its more detailed surface geometry. In comparison with V2, X-Toon produced a softer and more controllable stylised appearance by allowing the stylisation level to be adjusted independently of the illumination response, and was consequently retained as the toon shading component in all later technique combinations.

Despite this additional control, X-Toon remains a shading-based technique and therefore does not introduce edge information for interior features. Facial contours, clothing folds, and texturally encoded details thus remain visually unmarked unless they produce sufficient variation in the tonal response. This limitation is inherited from V2 and motivates the edge-based techniques introduced in V3.

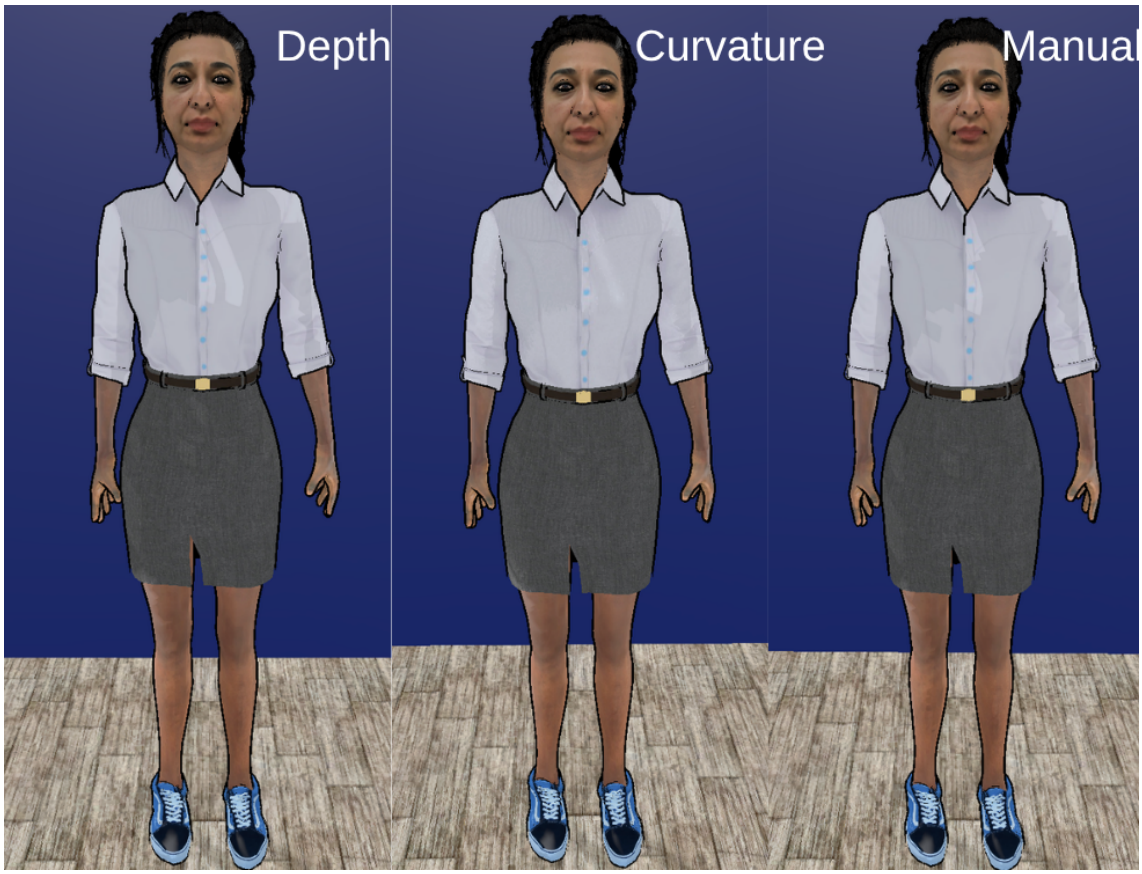


(a) Mixamo: depth, curvature, and manual abstraction modes.



(b) Meta SDK: depth, curvature, and manual abstraction modes.

Figure 4.9: V2.2 (X-Toon) across the three avatar platforms. From left to right: depth-driven, curvature-driven, and manually controlled abstraction.



(c) Avaturn: depth, curvature, and manual abstraction modes.

4.3.5 V3 — Screen Space Normal Edge Detection

Silhouette extraction techniques can be classified into image space, object space, and hybrid approaches [8]. Unlike the object space outline method introduced in V1, image space techniques analyse per pixel information generated during rendering to identify visually significant boundaries. One commonly used cue is the surface normal, whose orientation changes abruptly at creases, folds, and other geometric features. In prior work, per-pixel surface normals stored in G-buffer channels have been used as a cue for edge detection [34]. Discontinuities across these same channels detect silhouette, border, and crease edges in real-time NPR rendering, and crease edges in particular are recovered from the normal buffer [35]. It therefore gives image space techniques access to the interior geometric structure that silhouette methods alone leave undetected.

V3 builds on this image-space principle and introduces the first interior edge signal into the proposed rendering pipeline. It detects changes in the interpolated world space surface normal across neighbouring screen pixels, emphasising geometric contours alongside the silhouette outline rather than relying solely on tonal variation. This detection is realised through GPU screen-space derivative functions that evaluate the difference between adjacent fragments directly within the fragment pass. Consequently, they require neither a stored normal texture nor an additional rendering pass.

The GPU derivative operators compute the horizontal and vertical gradients of the world space surface normal $\mathbf{N}$, and the magnitude of the resulting gradient, E_N, is defined as:

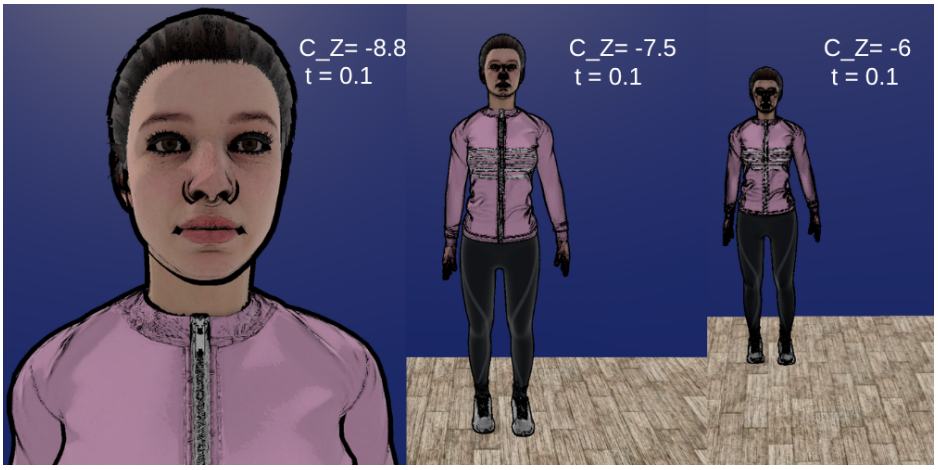
$$E_N = \sqrt{\left\| \frac{\partial \mathbf{N}}{\partial x} \right\|^2 + \left\| \frac{\partial \mathbf{N}}{\partial y} \right\|^2} \tag{4.4}$$

The gradient magnitude is high where the surface orientation changes abruptly, such as along geometric creases and clothing folds, and remains close to zero over smooth surfaces. A smoothstep function then converts this raw magnitude into a continuous edge response, preventing abrupt temporal changes as animated surfaces cross the detection threshold. The resulting edge mask is blended with the X-Toon shaded surface using the same compositing formulation adopted throughout the remaining edge detection techniques.

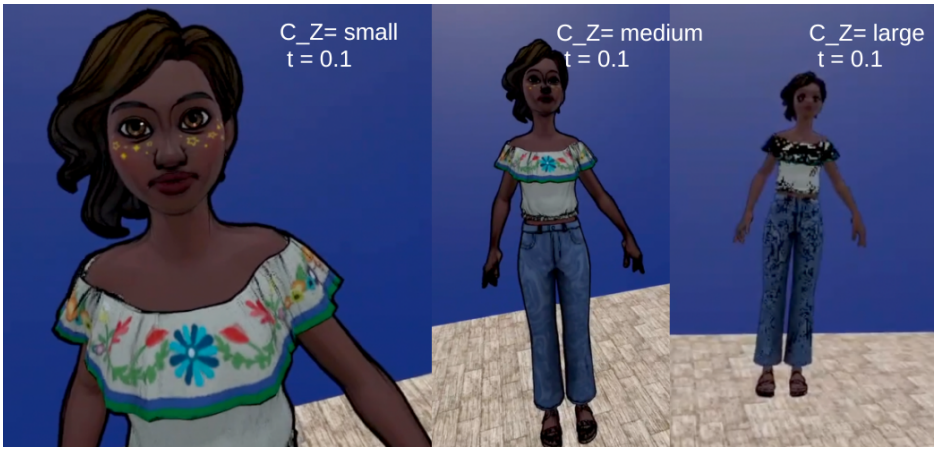
Across all three platforms, V3 complements the silhouette outline established in V1 by introducing interior contours derived from surface geometry. Figure 4.10 illustrates this across the three avatar platforms, where the edge output varies with both the normal information each platform exposes and the viewing distance. Mixamo and Avaturn both compute derivatives from shading normals, producing a rich edge network that improves the readability of clothing folds, seams, and other regions with pronounced surface curvature. The Meta Avatars SDK, by contrast, exposes only interpolated geometric normals through the shader hook. This restricted input results in considerably sparser interior contours, most noticeably on clothing surfaces where finer surface curvature would otherwise be detected.

Despite these improvements, the technique has two important limitations. First, because the normal gradient is evaluated in screen space, its magnitude depends on image resolution rather than object scale. As the viewing distance increases, geometric features span fewer screen pixels, causing progressively thicker interior edges. Second, the method responds only to changes in surface orientation, so texture boundaries and colour transitions that do not coincide with a normal discontinuity remain undetected.

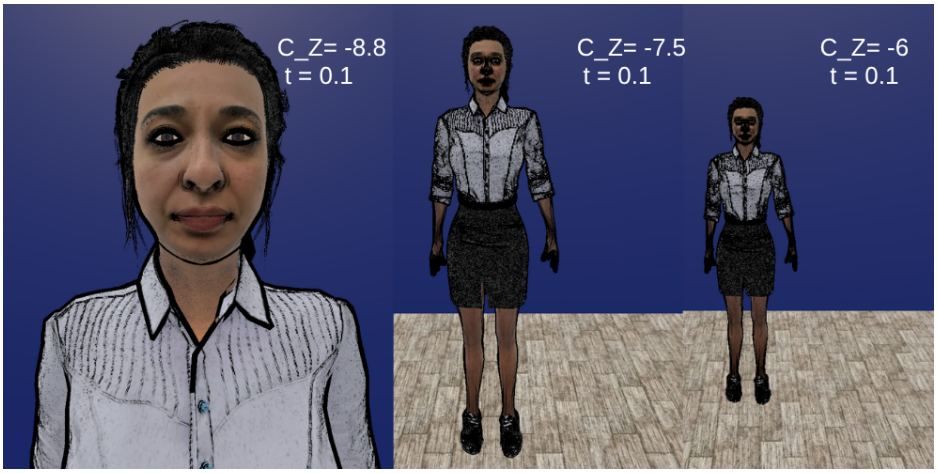
These limitations prevented V3 from being selected for the final rendering pipeline. Edge thickness scales with the viewing distance, producing inconsistent results. Concurrently, the reliance on surface orientation leaves many perceptually important facial and material boundaries unmarked because they are defined primarily by texture rather than geometry. Addressing these limitations required an edge detector sensitive to colour boundaries rather than surface orientation, motivating the Sobel-based approach introduced in V4.



(a) Mixamo: normal edge detection at increasing viewing distances.



(b) Meta Avatars SDK: normal edge detection at increasing viewing distances.



(c) Avaturn: normal edge detection at increasing viewing distances.

Figure 4.10: V3 (Screen Space Normal Edge Detection) across the three avatar platforms. C_Z: camera-to-avatar viewing distance; t: detection threshold.

4.3.6 V4 — Sobel Edge Detection

The Sobel operator is a gradient-based edge detector and one of the most commonly applied methods in digital image processing [11]. V4 applies the Sobel operator directly to the diffuse albedo texture rather than the surface normal field. This targets colour boundaries in the surface appearance, including clothing patterns, facial details, and painted surface structures, while avoiding the edge thickness growth with viewing distance observed in V3.

The Sobel operator estimates horizontal and vertical image gradients by convolving the input image with two 3×3 kernels [11]:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{4.5}$$

The weight of 2 assigned to the center row and column of each kernel provides a degree of spatial averaging. This gives the Sobel operator stronger noise rejection than simpler first-difference operators [11]. These two responses are combined to form the gradient magnitude

$$M_S(x, y) = \sqrt{G_x^2(x, y) + G_y^2(x, y)} \tag{4.6}$$

where G_x and G_y denote the Sobel responses in the horizontal and vertical directions [11]. Pixels for which M_S exceeds a configurable threshold are classified as edge pixels. They are then composited over the X-Toon shaded surface to produce the final stylised appearance.

Across the three avatar platforms, however, this output quality depends strongly on the available texture source. On Mixamo, the relatively clean painted albedo enables the detection of clothing boundaries and texture transitions with limited interference from unrelated intensity variations. In contrast, Avaturn uses photographic textures that contain illumination information captured during the reconstruction process. Consequently, the Sobel operator responds not only to material boundaries but also to baked lighting variations. As shown in Figure 4.11, this effect is particularly visible around the nose bridge, where the transition between highlight and shadow produces a strong response despite not representing a meaningful texture boundary. On the Meta Avatars SDK, the Sobel operator was applied to the available composited rendering output rather than the original diffuse texture. Although this introduces responses tied to scene lighting, the rendered texture contains less high-frequency photographic variation. Edge structures are thus cleaner and more consistent than on Avaturn, particularly along clothing boundaries.

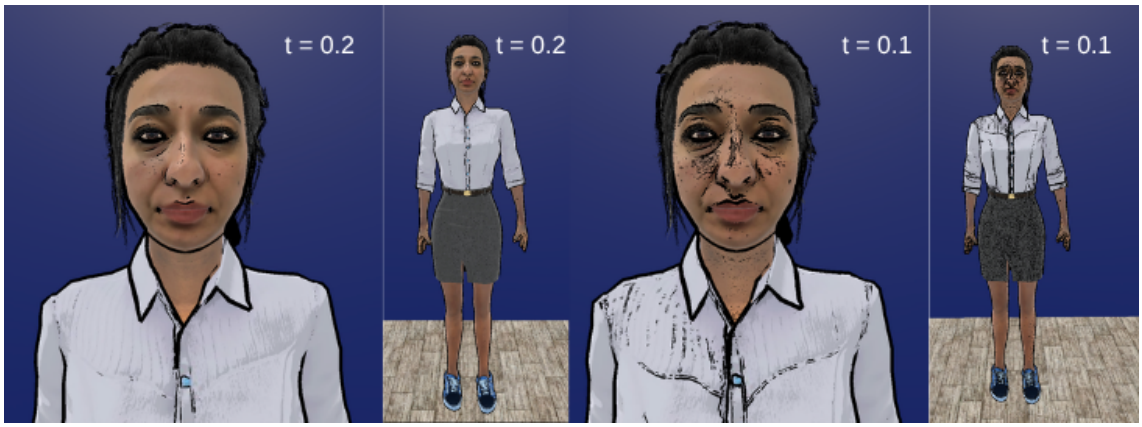
These input differences directly affect the threshold calibration required on each platform. Mixamo produced almost no unwanted edge responses at $t = 0.2$, whereas Avaturn yielded markedly more artefacts at the same threshold. This arose from its photographic texture content. Increasing the threshold on Avaturn suppressed these responses, but it also removed meaningful facial contour lines and eliminated segment boundaries

present at the lower setting, as Figure 4.12 shows. The Meta Avatars SDK requires lower threshold values because its composited output produces weaker gradients at material boundaries. This same property reduces sensitivity to fine texture variations and results in comparatively cleaner edge patterns. These observations confirmed that the Sobel response depends on the available appearance representation rather than solely on underlying geometry. The threshold must therefore be calibrated to each platform's signal characteristics rather than treated as a fixed operator property.

(a) Mixamo: Sobel edge detection at two threshold values.

(b) Meta Avatars SDK: Sobel edge detection at two threshold values.

Figure 4.11: V4 (Sobel Edge Detection) across the three avatar platforms. t: Sobel threshold.



(c) Avaturn: Sobel edge detection at the baseline threshold ($t = 0.2$).

Compared with V3, Sobel edge detection successfully captures texture-defined boundaries that remain invisible to normal-based methods, including clothing patterns, facial details,

and painted surface structures. The operator responds to all local intensity variations, however. Consequently, fine texture details, photographic illumination, and compression artefacts introduce unwanted responses alongside genuine structural boundaries. A further source of false detections arises from UV seam boundaries: neighbouring pixels on opposite sides of a seam correspond to distant locations in texture space, and the Sobel operator therefore interprets these discontinuities as strong edges. Both categories of false detection motivate V4.2, where Gaussian smoothing is introduced before gradient computation to reduce sensitivity to high-frequency variation.

Figure 4.12: Avaturn: Sobel edge detection at an increased threshold. Noise is reduced compared with Figure 4.11c, but meaningful facial contour lines are also eliminated.

4.3.7 V4.2 — Gaussian Prefiltered Sobel

The Sobel operator computes image gradients through partial derivatives. As this differentiation amplifies noise alongside genuine intensity changes, a preliminary noise suppression step is necessary to stabilise the edge response. Gradient operators therefore benefit from Gaussian prefiltering, since the Gaussian function is smooth in both spatial and frequency domains, reducing variation at fine spatial scales without the ringing artefacts produced by box averaging filters [11], [12]. Canny showed that the optimal 1-D step edge detector under additive noise conditions takes the form of the first derivative of a Gaussian kernel, which is mathematically equivalent to Gaussian smoothing followed by differentiation [27]. On this basis, V4.2 applies a discrete Gaussian filter before the Sobel operator, approximating this theoretically grounded detection strategy.

A discrete 3×3 Gaussian kernel is applied to the texture before the Sobel operator computes the image gradients:

$$G = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \tag{4.7}$$

Gaussian prefiltering produced cleaner edge maps than V4 across all three platforms, though the degree of improvement differed markedly by texture source. On Mixamo,

Gaussian smoothing reduced edge responses caused by compression artefacts and subtle texture variation while preserving the major clothing boundaries and facial features. On the Meta Avatars SDK, prefiltering similarly attenuated high-frequency artefacts in the composited rendering output, producing cleaner edge structures, particularly on clothing regions. Avaturn remained the most challenging platform because its photographic albedo contained baked illumination that introduced strong intensity gradients unrelated to surface structure. Although Gaussian smoothing attenuated some of these responses, many illumination gradients remained sufficiently strong to influence the Sobel output.

Overall, Gaussian prefiltering produced cleaner edge maps compared with V4 by reducing unwanted texture responses. However, this improvement came at the cost of weaker structural contours, particularly along features with limited contrast. Because Gaussian smoothing attenuated both noise and genuine edges, reducing false detections suppressed some meaningful structural information. Consequently, a compromise between contour preservation and noise reduction remained unavoidable.

Among the evaluated platforms, Avaturn exhibited the weakest performance under both V4 and V4.2. The baked illumination present in the photographic texture prevented reliable separation of structural boundaries from lighting variations, even after Gaussian smoothing. These observations motivated the texture preprocessing pipeline described in Section 4.5, where the source texture was edited to reduce gradients introduced by illumination before edge extraction. Combining the preprocessed texture with Gaussian prefiltered Sobel produced the most reliable structural edge detection for Avaturn.

Although Gaussian prefiltering improves the robustness of texture-based edge detection, its effectiveness remains dependent on the quality of the input texture. The following techniques therefore investigate post-processing stylisation approaches that operate on the rendered image rather than directly on the texture, reducing their reliance on texture quality while providing greater control over the final stylised appearance.

(a) Mixamo: Gaussian prefiltered Sobel edge detection.

(b) Meta Avatars SDK: Gaussian prefiltered Sobel edge detection.

(c) Avaturn: Gaussian prefiltered Sobel edge detection.

Figure 4.13: V4.2 (Gaussian Prefiltered Sobel) across the three avatar platforms. t: Sobel threshold.

4.3.8 V5 — Hierarchical Edge Detection with Multiple Cues

V5 adopts a post-processing architecture that operates on the rendered image rather than on the source texture, addressing the reliance on source texture quality identified in V4.2. It applies hierarchical edge detection to combine three complementary cues into a unified boundary representation: depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure, and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal. Each layer carries an independent weight, and the responses are blended through adjustable parameters so that the contribution of each cue can be controlled interactively [36].

Although the same hierarchical formulation was evaluated across all three avatar platforms, differences in the rendering architecture affected which edge cues could be com-

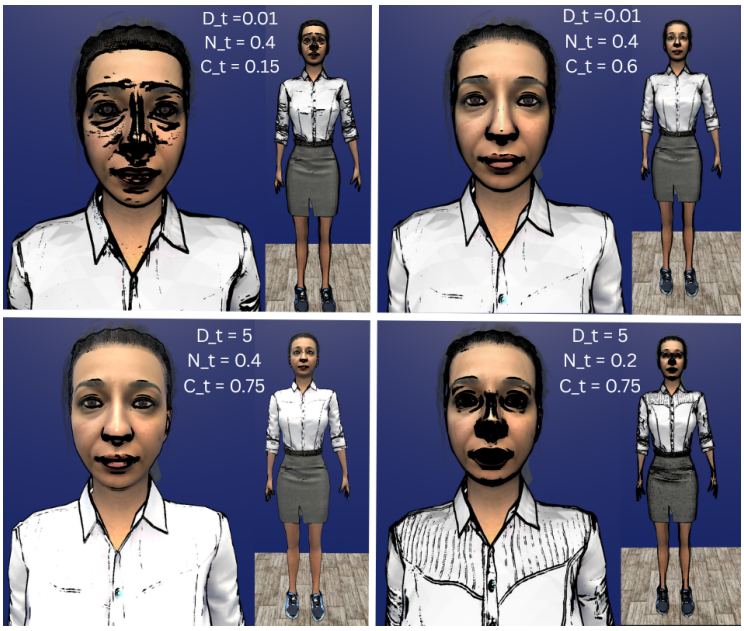
puted. On Mixamo and Avaturn, V5 operated as a post-processing effect that sampled the depth, normal, and colour buffers provided by the URP. By contrast, the Meta Avatars SDK restricts custom rendering logic to a per-fragment pass without access to global screen-space depth and normal buffers. Consequently, the implementation approximated depth from the fragment distance to the camera and combined only normal and texture cues, preventing reliable detection of the avatar silhouette against the background.

Figure 4.14 compares the results across the three avatar platforms. V5 detects a broader range of structural boundaries by combining multiple edge detection techniques on the Mixamo and Avaturn rendered frames. The depth threshold has little visual influence, since the inverted hull outline already provides the silhouette. The normal and colour thresholds instead control the density of interior contours. These contours trace geometric curvature, appearing along the underside of the nose and the outer lip edges, but do not produce clear boundaries between adjacent facial segments. Nonetheless, the colour gradient layer occasionally responds to illumination gradients in the rendered image, particularly around the nose bridge where shading transitions are interpreted as texture edges. On the Meta Avatars SDK, the absence of screen-space depth prevents silhouette detection, although combining the remaining cues still produces richer interior detail than the previous techniques. The composited SDK output contains baked lighting gradients absent from the raw albedo, making the texture layer more prone to false responses on this platform. An XDoG pre-blur applied before the colour gradient stage reduced these, as Figure 4.14(b) shows.

Compared with the previous techniques, V5 provides the most complete boundary representation by integrating silhouette, geometric, and texture information within a single framework. Nevertheless, because each layer contributes independently to the final edge map, the overall edge density increases and requires weight adjustment to avoid visually cluttered results. Moreover, the contribution of each layer remains spatially uniform across the rendered image, regardless of the local image content. This motivated the exploration of region-based stylisation in V6, where the surface appearance is reorganised into coherent colour regions rather than extended with additional edge marks.

(a) Mixamo: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

(b) Meta Avatars SDK: hierarchical edge detection at varying threshold combinations with XDoG pre-blur enabled and disabled.



(c) Avaturn: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

Figure 4.14: V5 (Hierarchical Edge Detection with Multiple Cues) across the three avatar platforms. N_t: normal threshold; C_t: colour threshold. D_t is not varied as a display parameter because the inverted hull outline already provides the silhouette boundary on Mixamo and Avaturn regardless of the depth threshold value, and the Meta Avatars SDK per-fragment pass has no access to the screen-space depth buffer.

4.3.9 V6 — Kuwahara Painterly Filter

The Kuwahara filter is a nonlinear smoothing operator that replaces each pixel with the mean colour of whichever overlapping neighbourhood subregion exhibits the lowest internal variance [37]. Subregions straddling a tonal boundary exhibit higher variance and are not selected, so uniform regions collapse into broad painterly fills while boundaries remain sharp. In its original formulation, however, the neighbourhood is partitioned into four fixed square subregions, making the output insensitive to local feature orientation and producing isotropic brushwork regardless of surface geometry. Kyprianidis et al. resolved this by computing a smoothed structure tensor and deriving an elliptical sampling window whose major axis aligns with the dominant direction of local image structure [13]. The resulting elliptical kernel causes marks to follow local surface curvature, rather than imposing the fixed rectangular grid of the isotropic variant.

V6 applies this filter to the rendered avatar surface to introduce a form of abstraction qualitatively different from the contour techniques of V3 through V5. Where those techniques add stylised marks over a photorealistic surface, the Kuwahara filter reorganises the surface appearance itself into coherent colour regions, producing an effect analogous to a painted or illustrated representation. A post-processing architecture is adopted on Mixamo and Avaturn because the anisotropic variant requires multiple rendering passes, which the Meta Avatars SDK single-pass constraint does not permit. On the SDK platform, Gaussian smoothing of the structure tensor requires a separate render pass and cannot be

performed within the hook. The tensor is instead estimated from screen-space derivatives at individual pixels, producing noisy orientation estimates.

Figure 4.15 compares the results across the three avatar platforms. On Mixamo and Avaturn, the full anisotropic implementation produced coherent painterly regions with well-oriented brushstrokes that follow the contours of clothing and facial features. On the Meta Avatars SDK, the noisy tensor causes the inverse-variance weighting to favour the most uniform colour patch in each neighbourhood rather than following feature orientation, amplifying local colour saturation instead of producing coherent painterly regions.

Achieving a clearly painterly effect, however, comes at a cost: at a kernel radius large enough for the painted regions to be perceptually distinct, fine facial features such as eyebrows, lip boundaries, and iris details tend to be subsumed into the surrounding skin region, reducing legibility. Reducing the kernel radius preserves these details but diminishes the painted quality of the result.

This trade-off between abstraction strength and feature legibility suggested that the Kuwahara filter is most effective when combined with a contour detection layer that restores structural boundaries.

(a) Mixamo: standard render (left) and Kuwahara painterly filter (right).

(b) Meta Avatars SDK: standard render (left) and Kuwahara painterly filter (right).

(c) Avaturn: standard render (left) and Kuwahara painterly filter (right).

Figure 4.15: V6 (Kuwahara Painterly Filter) across the three avatar platforms.

Figure 4.16: V7 (Combination: Kuwahara and Hierarchical) across the three avatar platforms. The Kuwahara filter produces coherent painterly regions; the hierarchical edge layer adds silhouette and crease contours. The Meta SDK variant omits the Sobel stage; see text.

4.4 Platform and NPR Technique Selection for the User Study

Because it is the only platform evaluated in Section 4.2 that provides photorealistic facial reconstruction, full blendshape animation, and unrestricted URP shader access, Avaturn was selected for the user study. Consequently, it stands as the sole candidate satisfying all requirements for a valid perceptual trust comparison.

Five components were considered as candidates for the NPR composite used in the user study condition C3: an inverted hull outline, Sobel edge detection, XToon tonal shading, a Kuwahara painterly post-processing pass, and a hierarchical depth edge layer. Parameter settings for all five, and whether the two post-processing passes should be retained at all, were determined through a pre-study ranking exercise, documented in Section 4.6. Following this exercise, the Kuwahara pass and the hierarchical depth edge layer were excluded from the final composite: participants ranked the unmodified configuration as significantly more trustworthy than every tested variant of each pass. The NPR condition for the user study is therefore realised through a composite of only the remaining three components: an inverted hull outline, Sobel edge detection, and XToon tonal shading. This arrangement satisfies the two requirements of the user study stimulus: facial contour legibility throughout animation and a clear tonal structure that conveys depth without photorealistic texture detail, while remaining perceptually distinct from the PBR baseline. The visual assessment procedure and full selection rationale are described in Chapter 6.

4.5 Avaturn Avatar Preprocessing

To reduce the edge detection artefacts caused by the reconstructed albedo texture established in Section 4.3.7, both the head mesh and the head albedo texture were preprocessed in Blender before the avatar was re-exported as a GLB file. Two operations were applied to the geometry and four to the texture.

For the geometry, Laplacian Smooth suppressed photogrammetric surface noise, removing pore and wrinkle geometry that would otherwise introduce false edge responses during the subsequent Sobel pass, with the neck seam pinned to prevent drift. Placed immediately below this step in the Blender modifier stack, a Subdivision Surface increased mesh density so that facial contours arrive at the fragment shader as a smooth silhouette boundary.

The four texture operations addressed two distinct problems in the head albedo: baked illumination variation that introduces spurious edge responses at boundaries unrelated to surface structure, and the absence of a strong tonal cue at the lip boundary. A Gaussian blur attenuated baked ambient occlusion, subsurface scattering, and pore detail from the photogrammetric albedo, suppressing the high-frequency intensity variation before Sobel detection [11]. A skin tone unification pass then blended skin pixels toward a common brightness and saturation baseline, collapsing the baked luminance falloff into a consistent tonal surface. The brow and eye socket regions were excluded from this step through UV masks derived from geometry vertex positions, since the chromatic contrast at those features is necessary for the Sobel operator to locate eyebrow and eyelid edges accurately. To address the second problem, saturation and brightness were enhanced across the mouth area, producing a vivid, uniform colour zone against the flattened skin. A lipline was then placed at the lip boundary by assigning a fixed dark red value to a programmatically detected paint stroke, providing a hard tonal step that Sobel detection and the X-Toon ramp in Unity can reliably identify.

Figure 4.17 shows the effect of the preprocessing pipeline and two NPR techniques applied to the modified avatar.

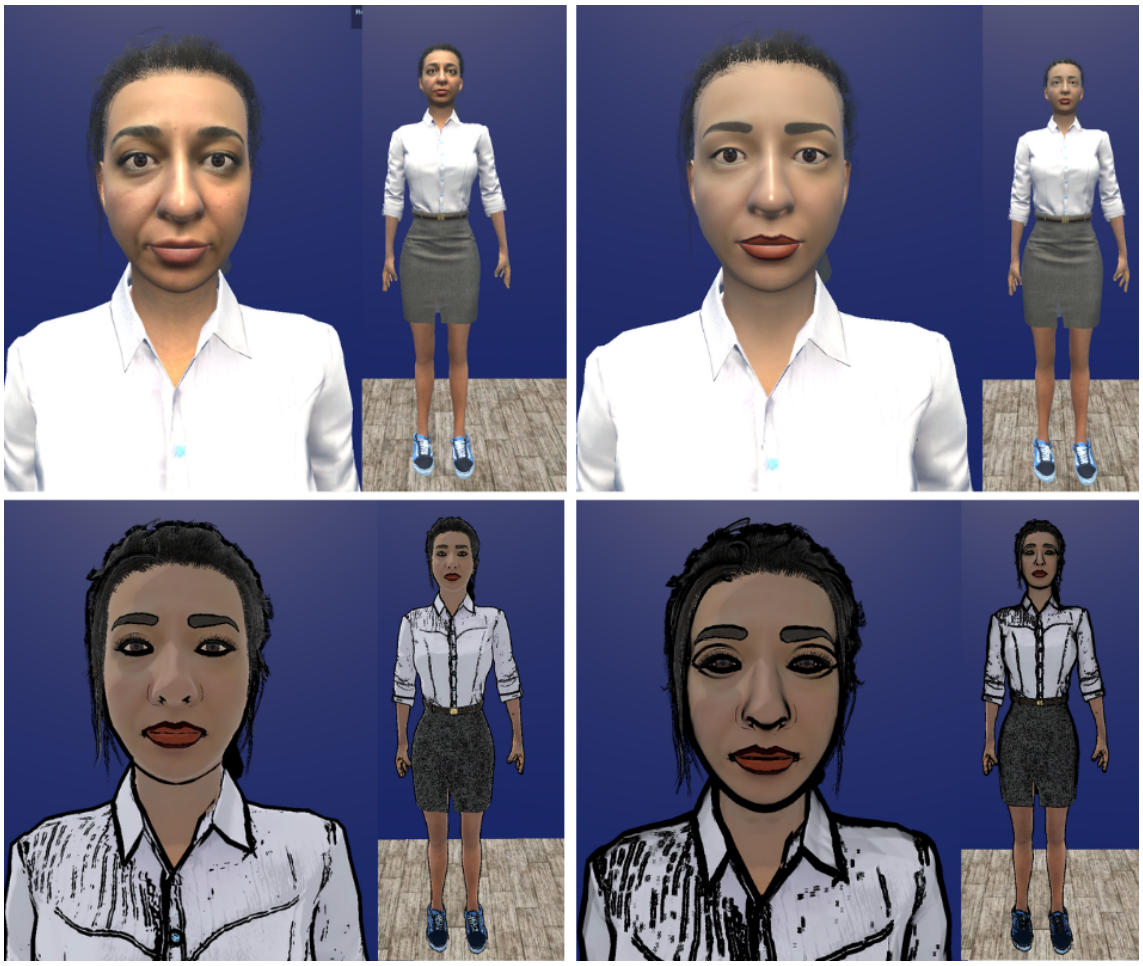


Figure 4.17: Avaturn avatar rendering progression. From left to right: (1) default Avaturn with no NPR processing; (2) Blender-preprocessed head texture abstraction only; (3) Gaussian prefiltered Sobel on the body with the customised Gaussian Sobel shader on the head, both combined with hierarchical edge detection post-processing; (4) hierarchical edge detection post-processing as the primary effect, also combined with the Gaussian Sobel pass. The two rightmost results include hierarchical post-processing in both cases.

4.6 Pre-Study Parameter Calibration

The candidate NPR composite for C3 combined five components: an inverted hull outline, Sobel edge detection, XToon tonal shading, a Kuwahara painterly filter, and a hierarchical depth edge layer. The outline establishes the silhouette, Sobel gradients identify interior contours at colour boundaries, and tonal shading controls colour quantisation. The two candidate post-processing passes each targeted a property the other three components could not provide: the Kuwahara filter would smooth texture regions into coherent painted areas, and the hierarchical layer would add depth-based boundary detection at geometry discontinuities where colour contrast is insufficient for Sobel. A Gaussian pre-filtering stage was available within the Sobel component but was disabled after preliminary testing showed that it attenuated meaningful edge responses alongside noise, offering no selective benefit at the parameter ranges relevant to this avatar. To determine the parameter settings for all five candidates and whether the two post-processing passes should be included at all, a ranking exercise was conducted with a pilot group before the main study, in which

participants ordered rendered images by perceived trustworthiness as an advisor. The exercise covered the parameter groups with the greatest expected influence on perceived edge density, tonal balance, or abstraction strength; parameters with comparatively minor expected visual impact, including rim light presence, Kuwahara kernel size, Gaussian blur strength, sector count, and normal edge strength, were instead set through direct visual inspection during development, since including every configurable parameter in the ranking exercise would have made the pilot study prohibitively long for participants.

The calibration covered five parameter groups, each selected because it exerts a direct influence on perceived edge density, tonal balance, or abstraction strength. In each round, one of these was varied while all remaining settings were held at the level fixed by the preceding test. Table 4.2 lists all stages and the values examined at each.

Table 4.2: Pre-study parameter calibration: the five parameter groups varied and the values tested in each round. Labels follow the convention used in Figure 4.18: R denotes the round number and the letter suffix (A, B, C, . . .) indexes each value in the order tested. R4A and R5A are identical to R1E: as the "none" baseline of their respective rounds, they carry forward the configuration selected at the end of Round 1.

Round	Parameter group	Values tested (A, B, C, . . . in order)
R1	Inner line threshold	0.01, 0.02, 0.03, 0.04, 0.05
R2	Outline width and depth offset	width: 0.003, 0.005, 0.007; offset: on / off
R3	Shadow strength and detail bias	strength: 0.0, 0.5, 1.0; bias: 0.0, 0.5, 1.0
R4	Kuwahara hardness and sharpness	none (= R1E); (18, 1); (9, 9); (1, 18)
R5	Hierarchical depth threshold	none (= R1E), 0.01, 0.02, 0.03, 0.04, 0.05

Forty-five pilot participants took part in the ranking exercise. Ballots that left a rank position blank or assigned the same rank to more than one image were excluded from that round's analysis, since both violate the forced full-ranking format; the number of valid ballots therefore varies slightly by round (between 40 and 44 of the 45 responses). Within each round, images were converted to Borda-count points (the lowest-ranked image receiving 1 point, the highest-ranked image receiving points equal to the number of options in that round) and summed across participants. An omnibus Friedman test established whether the ranking within a round differed from chance, and a pairwise Wilcoxon signed-rank test compared the top-ranked configuration against its runner-up, following the same nonparametric approach applied to the advisor-ranking measure in the main study (Section 4.7.4). Table 4.3 reports the outcome of each round.

Two outcomes determine the final composite. In R1, the numerically top-ranked threshold (0.03) was not statistically distinguished from the runner-up (0.05); it was retained regardless because the round's overall ranking was not uniform (the two lowest thresholds were significantly disfavoured), indicating the choice was not arbitrary even though the top two configurations were close. The same reasoning applies to R2, where the default outline width (0.005, depth offset on) was statistically indistinguishable from the narrower width (0.003, depth offset on); the default width was retained since no significant preference justified deviating from it. R3 produced no significant differences at all, so the untested midpoint configuration (shadow strength 0.5, detail bias 0.5) was retained as the reference condition. In both R4 and R5, however, the no-post-processing baseline was

Table 4.3: Outcome of the pre-study ranking exercise. The Friedman column reports the omnibus test across all configurations in a round; the Runner-up test column reports the pairwise Wilcoxon signed-rank test between the top-ranked configuration and its closest competitor.

Round	Top-ranked configuration	Friedman	Runner-up test
R1	threshold 0.03	$\chi^2_F(4) = 36.48$, $p < .001$	$p = .31$ vs. 0.05 (not distinguished)
R2	width 0.005, offset on	$\chi^2_F(5) = 126.19$, $p < .001$	$p = .82$ vs. width 0.003, offset on (not distinguished)
R3	strength 0.5, bias 0.5	$\chi^2_F(4) = 3.18$, $p = .53$	no configuration differed significantly
R4	none (no Kuwahara)	$\chi^2_F(3) = 38.97$, $p < .001$	$p = .003$ vs. best Kuwahara variant (significant)
R5	none (no hierarchical layer)	$\chi^2_F(5) = 117.20$, $p < .001$	$p < .001$ vs. best threshold (significant)

significantly preferred over every tested variant of the Kuwahara filter and the hierarchical depth layer respectively. Because trustworthiness is the primary outcome measure of the main study, both passes were excluded from the C3 composite on this basis, consistent with Section 4.4.

Figure 4.18 shows every rendered variant presented to participants across all five rounds, including the Kuwahara and hierarchical variants that were ultimately not retained. The proportion of participants who ranked each configuration first within its round is shown in Figure 1 in Section 0.

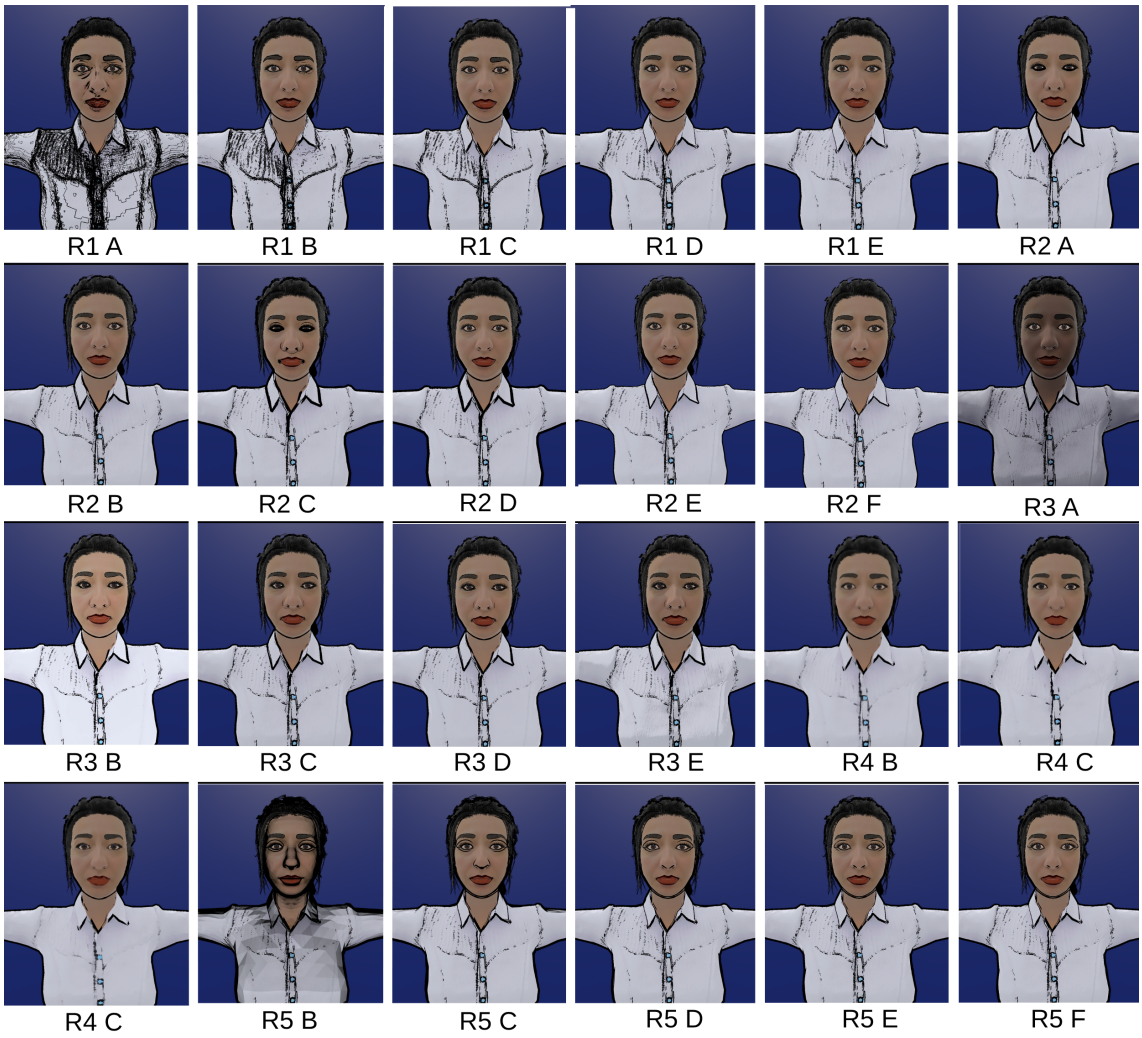


Figure 4.18: All rendered variants presented to participants during the pre-study parameter calibration. Each label follows the convention R*n*X, where *n* is the round number and X is the letter index of the value tested in that round (A being the first value, B the second, and so on, in the order listed in Table 4.2). R4A and R5A are not shown as distinct images: both are identical to R1E, the baseline configuration carried forward from Round 1 as the "none" starting point for Rounds 4 and 5.

4.7 User Study Design

4.7.1 Objective and Hypotheses

The user study addresses RQ3 by measuring how the visual representation of a speaker affects the perceived trustworthiness of their advice in a concrete decision situation. Three presentation conditions are compared: a real human speaker (C1), a photorealistic Avaturn avatar (C2), and the same Avaturn avatar rendered with NPR stylisation (C3). Five hypotheses are tested.

H1: The NPR avatar (C3) will produce lower eeriness ratings than the photorealistic avatar (C2). This follows from the Uncanny Valley hypothesis [3]: a photorealistic avatar at near-human fidelity occupies the most problematic region of the affinity curve, whereas

stylised abstraction deliberately relocates the representation to the stable, comfortable slope below the valley.

H2: The NPR avatar (C3) will not produce significantly lower self-reported trust than the photorealistic avatar (C2); a null effect is expected on the trust subscales. This is motivated by Canales et al., who found equivalent self-reported trust across three stylisation levels for a virtual doctor advisor [5], and by Gao et al., who found comparable trust between realistic and stylised agents [20].

H3 (exploratory): Both avatar conditions (C2 and C3) will produce lower self-reported trust than the real-person baseline (C1). Since the avatar was generated from the same individual appearing in C1, face identity and demographic appearance are held constant; any trust gap consequently describes the perceptual cost of computer-generated rendering rather than identity dissimilarity. Human speakers are generally expected to retain a trust advantage over computer-generated representations [4], but this comparison is treated as exploratory because animation naturalness is not controlled between C1 and the avatar conditions.

H4: Eeriness will negatively predict trust ratings across all conditions. This hypothesis is supported by the systematic review of Alipour et al., who report a direct negative eeriness-to-trust relationship in the majority of studies that tested it [18], and by the theoretical proposition of Ho and MacDorman that eeriness constitutes a specific affective response distinct from perceived human-likeness that interferes with positive social judgments [22].

H5 (exploratory): Anthropomorphism will mediate the relationship between condition and eeriness. Ho and MacDorman established that eeriness and human-likeness are orthogonal perceptual dimensions (reported $r \approx .02$), making this mediation path empirically testable rather than structurally guaranteed [22].

4.7.2 Conditions

The study compares three presentation conditions:

C1 — Real Person: A video recording of the same human speaker whose photograph was used to generate the Avaturn avatar, so that face identity, voice, and demographic appearance are held constant across all three conditions. C1 serves as the trust ceiling baseline against which both avatar conditions are compared.

C2 — Photorealistic Avatar: The same speaker scenario rendered using a photorealistic Avaturn avatar with standard PBR shading. Facial expressions are driven by blendshape animation to match the delivery timing.

C3 — NPR Avatar: The Avaturn avatar rendered with the NPR technique selected from the visual assessment as providing the best balance of abstraction and expressiveness.

Using the same Avaturn geometry for both C2 and C3 holds avatar identity, facial structure, and animation constant across the two avatar conditions, isolating render style as the sole independent variable; Latoschik et al. apply the same design principle to study the perceptual effects of avatar realism in social VR [26]. The theoretically critical comparison is C2 versus C3: because animation is held constant between these two conditions, any observed difference in trust or eeriness ratings is attributable to the change in rendering style alone.

4.7.3 Scenarios

Three video scenarios present naturalistic advice-seeking situations in which the speaker's trustworthiness is directly at stake. Each scenario has two plausible options and the speaker recommends one with explicit reasoning. Several design principles govern the scenario construction.

Ecological validity. The three scenarios are drawn from everyday situations that participants are likely to encounter in their social and professional lives: a career decision between two job offers, route planning in an unfamiliar city, and selection of a learning resource for career development. Grounding the scenarios in familiar real-world contexts ensures that trust judgments are directed at the advisor rather than spent interpreting an unfamiliar or artificial situation.

Vignette anchoring. Before each video, participants read a short context paragraph that anchors them in a specific, concrete situation with defined personal circumstances. Without this anchoring, trust ratings would reflect individual differences in life experience rather than responses to the speaker. The vignette eliminates this source of between-participant variance by placing every participant in the same role before each video begins.

Recommendation alternation. The three scenarios are designed so that the recommended option alternates across videos: Scenario 1 recommends Option A, Scenario 2 recommends Option B, and Scenario 3 recommends Option A. This prevents participants from detecting a systematic preference for either option label across the study.

Presentation order alternation. Within each script, the recommended option is not always discussed last. Discussing the recommended option last in every video would introduce a recency effect, where participants associate recommendation with position rather than with the speaker's reasoning. Scenario 1 and Scenario 3 discuss the recommended option first; Scenario 2 discusses the recommended option last. All scripts conclude with an explicit recommendation statement regardless of discussion order.

Script argumentative structure. Each spoken recommendation follows a fixed five-step structure, applied uniformly across all three scenarios: (1) state the recommendation explicitly; (2) acknowledge the primary drawback of the recommended option; (3) introduce the key advantage of the recommended option with an explicit contrastive marker; (4) name the central weakness of the non-recommended option; and (5) close with a concise restatement. This structure ensures the advisor engages visibly with trade-offs before

concluding, which holds rhetorical strategy constant across scenarios. Any difference in trust ratings between conditions therefore reflects the visual presentation of the speaker rather than variation in argumentation pattern between scenarios.

The three scenarios and their vignette texts are as follows.

Scenario 1 — Career Choice. The participant is placed in the role of a recent graduate choosing between two job offers, with financial pressure and location held constant as confounds. The speaker recommends Job A, judging career flexibility to outweigh the higher salary offered by Job B. This scenario primarily activates competence trust.

Scenario 2 — Travel Route. The participant must reach a hotel for a time-sensitive meeting, choosing between a direct 50-minute route that arrives ten minutes before the meeting and a faster 20-minute route requiring one transfer that arrives forty minutes early. The speaker recommends the faster option with the transfer, arguing that its 40-minute buffer provides enough recovery time to outweigh the transfer risk, whereas the direct route leaves an insufficient margin. This scenario activates competence trust through risk-benefit reasoning and benevolence trust through prioritising the participant's ability to arrive on time.

Scenario 3 — Learning Resource. The participant chooses between a paid online course with a named certificate and a free tutorial series covering the same material. The speaker recommends the paid course on the grounds that a recruiter cannot verify a free tutorial series, making the recommendation rest on credential value rather than content superiority. This scenario primarily activates integrity trust.

Scenario content, option framing, and speaker delivery are held constant across conditions; only the visual presentation of the speaker changes between C1, C2, and C3. The three scenarios are counterbalanced across conditions using a balanced Latin square (Table 4.4) to control for scenario-condition order effects and scenario-condition assignment confounds. In SoSci Survey, video slots are labelled Video 1, Video 2, and Video 3 without revealing condition identity; the mapping from slot to condition is stored per group and applied during data analysis.

Table 4.4: Balanced Latin square counterbalancing scheme. Each row is one participant group. Each condition (C1 = real person, C2 = photorealistic avatar, C3 = NPR avatar) appears once per group across the three video slots. Each scenario appears equally often with each condition across all six groups.

Group	Video 1	Video 2	Video 3
G1	C1 / Scenario 1	C2 / Scenario 2	C3 / Scenario 3
G2	C1 / Scenario 2	C3 / Scenario 1	C2 / Scenario 3
G3	C2 / Scenario 1	C1 / Scenario 3	C3 / Scenario 2
G4	C2 / Scenario 3	C3 / Scenario 2	C1 / Scenario 1
G5	C3 / Scenario 2	C1 / Scenario 1	C2 / Scenario 3
G6	C3 / Scenario 3	C2 / Scenario 1	C1 / Scenario 2

4.7.4 Procedure and Measures

Procedure

The study is conducted entirely online via SoSci Survey. Each participant completes a consent and demographic questionnaire, then watches three video blocks in the order determined by their assigned Latin Square group. Each block consists of four sequential steps: participants read a vignette establishing the scenario situation; they record an initial option preference before the video plays (this pre-video baseline is used to compute preference shift as a secondary exploratory measure); they watch the approximately 22-second recommendation video; and they complete the per-block questionnaire. After all three blocks, participants complete a post-study section. The total estimated completion time is approximately 20–25 minutes.

Per-Block Questionnaire

The per-block questionnaire is administered after each video in the fixed order Part A, Part C, Part D, Part B, Parts E–H. A scale-change separator and re-instruction mark the transition from the five-point semantic differential (Parts B–D) to the seven-point Likert (Parts E–H).

Part A — Comprehension check. A single multiple-choice item verifying that the participant watched the video and correctly identified the advisor's recommendation. Participants who select the wrong option or indicate they did not watch the full video are excluded from analysis.

Parts C, D, B — Semantic differential (5-point). These three subscales use a five-point semantic differential format and are presented in the order C, D, B. This order places the affective subscales (likeability and human-likeness) before the eeriness subscale to prevent uncanny valley priming from inflating eeriness ratings before trust is measured.

Part C assesses likeability using the five-item Godspeed Likeability subscale [21], with anchors ranging from *Dislike* to *Like*, *Unfriendly* to *Friendly*, and three further pairs. Part D assesses perceived human-likeness using the five-item Godspeed Anthropomorphism subscale [21], with anchors ranging from *Machinelike* to *Humanlike* and *Artificial* to *Lifelike*. Part B assesses eeriness using six items adapted from Ho and MacDorman [22], targeting strangeness, creepiness, unsettlingness, and perceptual discomfort; the instrument distinguishes eeriness from human-likeness as orthogonal perceptual dimensions, allowing H4 and H5 to be tested independently.

Parts E, F, G — Trust battery (7-point Likert). The eleven trust items are structured around the three-component Ability-Benevolence-Integrity (ABI) model of Mayer et al. [16], operationalised within the trusting-beliefs framework of McKnight and Chervany [17]. Part E contains four Competence items assessing perceived capability and expertise. Part F contains four Benevolence items assessing whether the advisor acts in the participant's interest. Part G contains three Integrity items assessing transparency and honesty. All items use a seven-point response format (1 = Strongly Disagree, 7 = Strongly Agree).

Measuring the three components separately allows differential effects of rendering style on the distinct facets of trustworthiness to be detected; a composite single score would mask such effects [17].

Part H — Response to recommendation (7-point Likert). Two custom items: H1 measures agreement with the advisor's recommendation; H2 measures perceived warmth and approachability of the advisor. H2 is grounded in findings from Gao et al. that stylisation can affect perceived warmth independently of trust subscale ratings [20].

An embedded attention check item is included in each of the three blocks, with a different target response per block to prevent pattern detection. The open-ended reflection question (Part J) is included in Block 2 only, to avoid increasing response fatigue in Blocks 1 and 3.

Post-Study Section

After all three video blocks, participants complete a post-study section containing two types of measures. An advisor preference ranking question (P1) asks participants to rank all three advisors from most to least preferred to consult, without reference to any specific recommendation. This is the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al. [5]; it captures implicit trust judgment that explicit Likert ratings may not fully reflect, addressing the methodological gap identified by Alipour et al. [18]. A rank order is used rather than a forced choice because H3 predicts C1 dominance; a forced-choice format would leave insufficient variance to distinguish C2 from C3 in the behavioural measure. A visual style manipulation check (Part I) asks participants to rate the perceived visual stylisation of each advisor simultaneously; post-study placement prevents stylisation awareness from priming trust ratings between blocks.

A propensity-to-trust item (seven-point Likert: *In general, I tend to trust people I meet for the first time*) is included in the demographics section as an individual-level covariate, following Mayer et al. [16], who identify disposition to trust as a stable individual difference that should be controlled when comparing interpersonal trust ratings across conditions.

Analysis Plan

The primary analysis is a one-way repeated-measures MANOVA with condition (C1, C2, C3) as the within-subjects factor and the six subscale composite scores (B_score, C_score, D_score, E_score, F_score, G_score) as dependent variables. If the omnibus MANOVA is significant (Pillai's trace, $p < .05$), univariate follow-up ANOVAs are conducted per subscale, followed by pairwise comparisons with Bonferroni correction ($\alpha_{\text{corrected}} = .05/3 = .017$). Effect sizes are reported as partial η^2. The theoretically critical pairwise comparison is C2 versus C3. Because H2 predicts equivalence rather than a difference, a non-significant pairwise result alone provides insufficient evidence; a two one-sided tests (TOST) equivalence test with a pre-specified bound of Cohen's $d = 0.3$ is applied to the C2 versus C3 trust composite to provide positive evidence of negligible difference. For H4, person-centred multilevel regression is used, with within-participant

eeriness as the predictor of within-participant trust, isolating the eeriness-to-trust relationship from between-condition variance in both measures. H5 is addressed via mediation analysis with anthropomorphism score as the proposed mediator in the C2-to-C3 condition shift. For P1, a Friedman test compares mean advisor ranks across the three conditions, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha_{\text{corrected}} = .017$); the C2 vs C3 comparison is the critical pairwise test.

Sample Size

A power analysis conducted for a three-condition within-subjects ANOVA targets a medium effect size of partial $\eta_p^2 = 0.06$, $\alpha = .05$, and power $= 0.90$, yielding a minimum of $n = 38$ participants. The recruitment target is set to $n = 48$ (eight per Latin Square group) to accommodate anticipated attrition and exclusions from the comprehension check. A balanced sample of eight per group also ensures complete counterbalancing across all six condition-scenario assignment patterns.

4.7.5 Ethical Considerations

The study was reviewed and approved by the University of Konstanz ethics committee. Participants provided written informed consent before participation. Data is anonymised at collection; no personally identifiable information is stored alongside questionnaire responses. Participation was voluntary with the right to withdraw at any time.

5 Implementation

This chapter documents the technical realisation of the NPR avatar system. It covers the development environment, the integration path for each avatar platform, the implementation of the shared rendering structure, and the HLSL-level details of each stylisation technique developed to date. Runtime tooling developed to support iterative development is described at the end of the chapter.

5.1 Development Environment

The system was developed in Unity 6 (version 6000.0.60f1) using the Universal Render Pipeline (URP). The target hardware is the Meta Quest family of standalone VR headsets. The Meta Avatars SDK version 40.0.1 was used for the SDK-managed avatar platform. All shader code is written in HLSL and targets the URP forward rendering path.

5.2 Avatar Platform Integration

5.2.1 Mixamo

Mixamo characters are exported as FBX assets and imported into Unity as standard skinned mesh renderers. No proprietary shader system is involved, so all URP materials can be replaced directly with custom NPR shaders without restriction. The Mixamo character Jody served as the primary prototyping target during technique development, allowing the NPR shaders to be iterated freely before the more constrained platform integrations were addressed.

5.2.2 Meta Avatars SDK

The Meta Avatars SDK manages avatar materials through its Extensible Shader Framework (ESF), which generates platform-specific HLSL from parameterised templates. The generated source is not intended to be edited directly. Custom logic is instead injected through named hook functions that the ESF exposes at defined points in the shader pipeline.

The hook used in this thesis is `AppSpecificPostManipulation`, which executes per fragment after all SDK PBR lighting — subsurface scattering, eye glints, hair shading — has been computed. It receives the accumulated lit colour and returns a modified colour, giving full control over the final fragment output without disrupting any SDK rendering logic. The hook is defined across three files that the ESF includes automatically during shader compilation:

- `app_functions.hlsl` — the body of `AppSpecificPostManipulation`; dispatches to the active NPR technique based on the `_NPREffect` keyword

- `app_declarations.hlsl` — uniform declarations for all NPR parameters, shared across techniques
- `app_variants.hlsl` — shader feature keyword declarations that activate each technique branch

A further complication is the SDK's LOD management system, which replaces avatar materials automatically as camera distance changes. This reverts any custom shader assignment made at load time. The `AvatarShaderSwapper.cs` script addresses this by running a periodic coroutine that re-applies the custom NPR materials after each LOD transition is detected. The script also handles the initial material replacement, waiting for the asynchronous avatar load to complete before copying base colour and normal textures from the SDK materials into the custom shader properties.

5.2.3 Avaturn

Avaturn exports avatars in the `.glb` format, the binary container of the glTF 2.0 standard. Unity does not natively support runtime glTF loading, so the GLTFast package is used to load the asset at runtime. On load completion, GLTFast produces a standard Unity GameObject hierarchy with skinned mesh renderers and PBR materials assigned from the glTF material definitions, including albedo, normal, and metallic-roughness textures. Because Avaturn avatars carry no proprietary shader constraints once imported, the custom NPR materials are assigned to the resulting skinned meshes using the same approach as Mixamo: direct material replacement after load. Blendshape animation targets exported by Avaturn are preserved through the GLTFast import and driven at runtime via `SkinnedMeshRenderer.SetBlendShapeWeight` for the video recording scenarios used in the user study. Facial animation is driven using Apple ARKit: the Live Link Face application streams all 52 ARKit blendshape coefficients and head rotation over UDP from an iPhone to Unity, where a custom component maps each coefficient to the corresponding Avaturn blendshape target.

5.2.4 Common Scene Environment for Cross-Avatar Comparison

A key requirement for meaningful cross-avatar comparison is that any visible differences between the Jody (Mixamo) and Avaturn avatars arise from the NPR technique itself, not from incidental differences in the rendering environment. To satisfy this, both avatars are placed in the same Unity scene and rendered under identical environmental conditions.

A single directional light is used as the primary illumination source for both avatars. Its transform — rotation, intensity, and colour temperature — is shared: the light GameObject exists once in the scene hierarchy and both avatar GameObjects fall within its influence. The light direction is set to a front-upper-left angle (-30° elevation, -30° azimuth from the avatar forward vector), which is a standard portrait lighting configuration that reveals facial structure and clothing wrinkles without casting extreme shadows. Intensity is fixed at 1.0 lux with a neutral white colour so that no colour bias is introduced by the light source.

Both avatars stand in the same physical room geometry. The environment is a studio-style enclosure with a neutral grey floor and walls, a controlled skybox, and no emissive surfaces, ensuring that indirect lighting contributions — sampled from spherical harmonics in the URP forward path — are identical for both avatars. Ambient intensity and the GI contribution are baked from the shared skybox, so the ambient component of the lighting equation is the same regardless of which avatar is being evaluated.

This configuration means that when a researcher or participant views the Jody and Avaturn avatars side by side or switches between them, all differences in shading, edge quality, and tonal response are attributable to the technique and its interaction with each avatar's specific mesh and texture assets, rather than to environmental factors. It also ensures that the scalar parameter values tuned for one avatar — outline width, toon band count, shadow strength — have the same perceptual effect on the other.

5.3 Shared Rendering Structure

On the Mixamo and Avaturn platforms, each NPR technique is a standalone multi-pass URP shader with an `NPROutline` pass and a `ForwardLit` pass; technique switching is performed by replacing the active material set. On the Meta SDK platform, all techniques share a single shader, `Avatar-Meta-UGB.shader`. The forward-lit dispatch lives in `Style2MetaAvatarCore.hlsl`, which reads the active `_NPREffect` keyword and branches to the corresponding technique include file. Switching techniques at runtime requires only toggling the keyword via `Material.EnableKeyword` and `Material.DisableKeyword` without recompilation.

5.3.1 Inverted Hull Outline Pass

The outline pass implements the inverted hull silhouette. In the Mixamo and Avaturn shaders the pass sets `Cull Front`, `ZWrite On`, `ZTest Less`; in the Meta SDK shader it sets `Cull Front`, `ZWrite Off`, `ZTest LEqual` to avoid conflicting with the SDK's depth compositing. The vertex shader displaces each vertex outward along its world-space normal by `_OuterOutlineWidth` (Mixamo/Avaturn, world-space metres) or `_OutlineWidth * 0.001` (Meta, converted from UI units). The fragment outputs a flat `_OuterOutlineColor` or `_OutlineColor`. An `_OutlineDepthBias` property shifts the clip-space Z coordinate toward the camera to prevent z-fighting where the expanded shell and the original surface meet at shallow angles.

5.3.2 Alpha Cutout Handling

Alpha cutout is controlled by the `_EnableAlphaTest` flag rather than being always-on. When enabled, both the outline pass and the forward-lit pass sample the base colour texture alpha channel and call `clip(alpha - threshold)` before any displacement or shading is applied. In the outline pass this ensures the expanded hull is discarded wherever the hair card texture is transparent, preventing a rectangular border from appearing around each card. The cutout threshold is exposed in the runtime UI and

adjusted per avatar submesh — a lower threshold is typically used for eyelashes (dense geometry) than for hair cards (sparse, large transparent regions).

This explicit alpha test is necessary on the Mixamo and Avaturn platforms because their outline pass is a minimal, self-contained `Cull Front` pass with no knowledge of the surface material's transparency.

The Meta Avatar SDK outline pass does not require this mechanism. The outline is implemented as a named pass inside the same multi-pass shader that handles the surface, using the same `Style2MetaAvatarCore.hlsl` include as the lit pass. The SDK's fragment pipeline — including its own internal alpha handling for hair and eyelash geometry — runs to completion before the `AppSpecificPostManipulation` hook is invoked. Any fragment that the SDK would discard as transparent is already clipped before the hook's `o.color = _OutlineColor` line is reached. Alpha cutout is therefore subsumed by the SDK and requires no additional shader property or UI control on the Meta platform.

Visual results comparing alpha test disabled and enabled are documented in Figure 4.5 in Chapter 4.

5.4 NPR Technique Implementation

All shader parameters described in the following subsections are accessible through two routes. The primary route is the Unity Inspector: each parameter appears as a labelled property field on the material asset and can be adjusted directly in the Editor during offline development. The secondary route is the in-VR WorldSpace UI panel described in Section 5.5.1, which exposes the same properties as interactive sliders and toggles inside the running application on the headset. Both routes write to the same underlying `MaterialPropertyBlock` values; changes made through either interface take effect immediately without shader recompilation. Technique switching — enabling or disabling an NPR variant — is performed through a dropdown row in the in-VR panel, which calls `Material.EnableKeyword` and `Material.DisableKeyword` on the active material set.

5.4.1 V2 — Toon Shading

Three separate shader implementations exist for V2, one per avatar platform. The outline pass is structurally identical for Mixamo and Avaturn (inverted hull, normal-offset displacement, explicit alpha cutout); the Meta variant shares the same inverted hull geometry but omits the explicit alpha test because the SDK's fragment pipeline handles hair and eyelash transparency internally before the outline hook fires (see Section 5.3.2). The toon shading pass differs because the Mixamo and Avaturn variants operate on raw $\mathbf{N} \cdot \mathbf{L}$, while the Meta variant operates on the composited PBR output.

Mixamo

Shader: `Assets/Shaders/JadeNPRShaders/V1_ToonShading_GeometryOutline.shader`
Pipeline target: URP, `#pragma target 3.0`

The toon pass quantises the Lambertian dot product $\mathbf{N} \cdot \mathbf{L}$ per step using a per-band smoothstep. The expression after unification with the Avaturn version is:

```
float steps  = max(1.0, _ToonSteps);
float scaled = NdotL * steps;
float band   = floor(scaled);
float frac   = scaled - band;
float blend  = smoothstep(1.0 - _ToonSmoothness, 1.0, frac);
float toon   = saturate((band + blend) / steps);
toon = lerp(1.0 - _ShadowStrength, 1.0, toon);
```

Listing 5.1: V2 toon quantisation : Mixamo and Avaturn (unified).

Each band boundary receives an independent narrow blend of width `_ToonSmoothness`, so normal jitter near any seam produces a localised crossfade rather than snapping the whole surface patch between bands. An additive ambient `_AmbientColor` and a rim term `pow(1 - NdotV, RimPower)` are added to the quantised diffuse.

Avaturn

Shader: `Assets/Shaders/AvaturnNPRShaders/V1_ToonShading_GeometryOutline.shader`
Pipeline target: URP, `#pragma target 3.5`
Materials: six materials covering body, head, hair, eyelash, look, and shoe submeshes

The toon pass is identical to Mixamo. Both vertex passes include a conditional `OVR_FETCH_POS_NORM` call that resolves to a no-op for standard GLTFast avatars; the bridge is retained through all subsequent technique versions for consistency. All six materials share the same parameter set via the `AvaturnSlotPresets` editor (`_ToonSteps` = 4, `_ToonSmoothness` = 0.05, `_ShadowStrength` = 0.5, `_OuterOutlineWidth` = 0.005 m, `_RimPower` = 4). The sole per-submesh variation is alpha cutout: hair and eyelash materials set `_EnableAlphaTest` = 1 with `_AlphaCutoff` = 0.1 and activate `_UseOutlineDepthOffset`; the remaining submeshes leave alpha testing disabled.

Meta Avatars SDK

Shader: `Assets/Shaders/MetaAvatarShaders/Avatar-Meta-UGB.shader`
Keyword: `EFFECT_TOON` (enabled via `ENABLE_NPR_EDGES`)
Technique file: `Assets/Shaders/MetaAvatarShaders/NPREffect_Toon.cginc`

The Meta implementation runs inside `AppSpecificPostManipulation`, which receives the fully composited PBR colour produced by the SDK's `STYLE_2_STANDARD` renderer. The quantisation therefore operates on RGB directly, preserving hue while stepping luminance:

```
float  bands      = max(2.0, _ToonColorBands);
float3 posterized = floor(color.rgb * bands + 0.5) / bands;
color.rgb = lerp(color.rgb, posterized, _ToonPosterizeStrength);

float lum  = dot(color.rgb, float3(0.2126, 0.7152, 0.0722));
```

```
color.rgb  = lerp(float3(lum, lum, lum), color.rgb, _ToonSaturation
    );
```

Listing 5.2: V2 posterisation : Meta SDK (`NPREffect_Toon.cginc`).

The `floor(rgb * bands + 0.5) / bands` expression rounds each channel to the nearest multiple of 1/bands (round-to-nearest rather than floor-to-band), which distributes quantisation error symmetrically and avoids the darkening bias of a plain floor. Saturation adjustment runs after posterisation to prevent clamping artefacts. Because the input already contains Meta's subsurface scattering, anisotropic hair, and eye glints, the posterised output stylises a physically correct image rather than reconstructing lighting from scratch.

The outline pass in this shader is a dedicated `NPROutline` pass with `Cull Front`, `ZWrite Off`, `ZTest LEqual`. The vertex shader displaces each SDK-skinned vertex outward by:

```
float3 worldPos = o.v_WorldPos + normalize(o.v_Normal) *
    _OutlineWidth * 0.001;
```

Listing 5.3: Outline vertex displacement : Meta SDK outline pass.

The `0.001` scale converts `_OutlineWidth` from millimetre-scale UI units to metre-scale world space. The outline is toggled via `_OutlineEnabled`; the fragment discards if the value is below 0.5. `ZWrite Off` is set as described in Section 5.3.1.

Table 5.1: V2 implementation comparison across the three avatar platforms.

	Mixamo	**Avaturn**	**Meta SDK**
Quantisation method	Per-step smoothstep, NdotL	Per-step smoothstep, NdotL	Round-to-nearest, composited RGB
Skinning bridge	None	No-op macro (GLTFast SkinnedMeshRenderer)	SDK native
Rim light	Additive, computed	Additive, computed	Pre-composited by Meta PBR
Outline depth write	`ZWrite On`	`ZWrite On`	`ZWrite Off`
Outline toggleable	No	No	Yes (`_OutlineEnabled`)
Shadow source	None (`GetMainLight` without shadow coords)	None (same)	Pre-composited by Meta PBR

Visual results for V2 across all three platforms are documented in Figure 4.7 in Chapter 4.

5.4.2 V2.2 — X-Toon Extended Toon Shading

Three shader implementations exist for V2.2, one per avatar platform. The outline pass is unchanged from V2 across all three; the forward-lit pass replaces the procedural quantisa-

tion of V2 with a 2D ramp texture lookup, adds an abstraction compression stage, and optionally applies Normal Field Abstraction on the Mixamo and Avaturn platforms [10].

Mixamo

Shader: `Assets/Shaders/JadeNPRShaders/XToon_2DRamp.shader`
Pipeline target: URP, `#pragma target 3.5`

The forward-lit pass begins by optionally modifying the surface normal via Normal Field Abstraction. When `_UseAbstractNormals` is enabled, an *abstract normal map* — a simplified or blurred version of the surface normal map — is sampled, transformed from tangent space to world space through the TBN matrix, and linearly interpolated with the geometry normal:

```
float3 abstractN  = UnpackNormal(SAMPLE_TEXTURE2D(
                        _AbstractNormalMap,
    sampler_AbstractNormalMap, uv));
float3x3 TBN      = float3x3(tangentWS, bitangentWS, normalWS);
float3 abstractWS = normalize(mul(abstractN, TBN));
normalWS          = normalize(lerp(normalWS, abstractWS,
    _NormalSmoothing));
```

Listing 5.4: Normal Field Abstraction : Mixamo and Avaturn.

When `_NormalSmoothing` is high, surface normals within a smooth region converge toward each other, causing the 2D ramp to return a consistent colour band across that region and producing a graphic, flattened read of the surface shape.

The U coordinate for the ramp is derived from the Lambertian dot product modulated by URP shadow attenuation, then remapped and interpolated toward the ramp midpoint to control light sensitivity:

```
Light mainLight = GetMainLight(input.shadowCoord);
float3 lightDir = normalize(mainLight.direction);
float NdotL     = dot(normalWS, lightDir) * mainLight.
    shadowAttenuation;
float rampU     = lerp(0.5, saturate(NdotL * 0.5 + 0.5),
    _LightSensitivity);
```

Listing 5.5: U axis (lighting) computation : Mixamo and Avaturn.

The V coordinate (abstraction level) is computed from the fragment's world-space camera distance in depth mode, from the screen-space normal derivative magnitude in curvature mode, or from a constant in manual mode, selected at compile time via `shader_feature_local` keywords. The ramp is then sampled at `(rampU, rampV)`, and an abstraction compression stage flattens the lighting bands as the abstraction level rises:

```
float abstractU    = lerp(rampU, 0.5, rampV * 0.6);
float dynSmoothing = lerp(_RampSmoothing, _RampSmoothing + 0.35,
    rampV);
float shadowMask   = smoothstep(0.5 - dynSmoothing,
```

```
                                    0.5 + dynSmoothing, abstractU);
float3 toonAlbedo    = albedo * rampColor;
float3 shadowedColor = lerp(toonAlbedo * _ShadowColor.rgb,
   toonAlbedo, shadowMask);
float3 finalColor    = lerp(albedo, shadowedColor, _ShadowStrength)
   ;
```

Listing 5.6: Abstraction compression and shadow mask : Mixamo and Avaturn.

As *V* increases, the U compression pulls rampU toward the midpoint while the smoothstep width widens, softening band boundaries rather than sharpening them. A Blinn-Phong specular highlight is added after the shadow stage; `_SpecularSmoothness` controls the transition width, with values near zero producing a sharp, ink-painted dot.

Avaturn

Shader: `Assets/Shaders/AvaturnNPRShaders/XToon_2DRamp.shader`
Pipeline target: URP, `#pragma target 3.5`
Materials: five materials covering body, head, hair, eyelash, and look submeshes

The Avaturn V2.2 shader shares the same NdotL-based U axis, abstraction compression, and Blinn-Phong specular stages as the Mixamo implementation, and similarly applies Normal Field Abstraction via TBN [10]. Both vertex passes include the SDK vertex fetch bridge, consistent with V2; for standard GLTFast meshes the bridge is a runtime no-op and adds no overhead.

The five per-submesh materials differ only in their texture assignments; all parameters — ramp texture, smoothing, abstraction mode, specular — are tuned uniformly across the set via the `AvaturnPresetShaderGUI` editor, which supports slot-based preset save and recall to maintain consistent looks across submeshes.

Meta Avatars SDK

Hook file: `Assets/Shaders/MetaAvatarShaders/NPREffect_XToon.cginc`
Keyword: `EFFECT_XTOON`
Entry point: `AppSpecificPostManipulation`

The Meta implementation runs inside `AppSpecificPostManipulation`. Unlike the Mixamo and Avaturn shaders, this hook does not own the vertex stage; world-space position is forwarded from the SDK vertex output as an explicit fifth argument to `ApplyNPREffect`. This makes the same U axis computation as the standalone platforms possible: `GetMainLight` is called with a per-fragment shadow coordinate derived from that world-space position, and the shadow-attenuated NdotL is computed using the same approach as on the standalone platforms:

```
float4 shadowCoord = TransformWorldToShadowCoord(positionWS);
Light mainLight    = GetMainLight(shadowCoord);
float3 lightDir    = normalize(mainLight.direction);
```

```
4 float NdotL        = dot(normalWS, lightDir) * mainLight.
    shadowAttenuation;
5 float rampU        = lerp(0.5, saturate(NdotL * 0.5 + 0.5),
    _XToonLightSensitivity);
```

Listing 5.7: U axis (shadow-attenuated NdotL) : Meta SDK (`NPREffect_XToon.cginc`).

The surface tangent frame is not available in the hook, so Normal Field Abstraction via an abstract normal map is not supported; only the geometric smoothing path is available on this platform. The abstraction compression and dynamic smoothing stages are applied identically to the standalone platforms. The V axis uses the same three-branch structure (Depth / Curvature / Manual), but selected via `multi_compile` instead of `shader_feature_local`, making all three variants available for runtime switching via the VR parameter panel. The same `lightDir` and shadow attenuation from the U axis computation are reused for the specular term.

Table 5.2: V2.2 implementation comparison across the three avatar platforms.

	Mixamo	**Avaturn**	**Meta SDK**
U axis source	NdotL, shadow-attenuated	NdotL, shadow-attenuated	NdotL, shadow-attenuated (per-fragment `GetMainLight`)
V axis computation	`shader_feature_local` keywords (Depth / Curvature / Manual)	`shader_feature_local` keywords	`multi_compile` keywords (same three modes, runtime-switchable)
Normal Abstraction	TBN abstract normal map or geometric smoothing	TBN abstract normal map or geometric smoothing	Geometric smoothing only (no TBN in hook)
Specular model	Blinn-Phong, shadow-attenuated	Blinn-Phong, shadow-attenuated	Blinn-Phong, shadow-attenuated (reuses `GetMainLight` result)
Skinning bridge	None	None (GLTFast SkinnedMeshRenderer)	SDK native
Shadow source	Per-vertex `GetShadowCoord`, interpolated	Per-vertex `GetShadowCoord`, interpolated	Per-fragment `TransformWorldToShadowCoord`

Visual results for V2.2 across all three platforms are documented in Figure 4.9 in Chapter 4.

5.4.3 V3 — Screen-Space Normal Edge Detection

Shaders: `Assets/Shaders/MetaAvatarShaders/NPREffect_NormalEdge.cginc` (Meta SDK); `Assets/Shaders/JadeNPRShaders/V2_NormalEdgeDetection.shader` (Mixamo); `Assets/Shaders/AvaturnNPR` (Avaturn)

Extra texture samples per fragment: 1 normal map sample (Mixamo / Avaturn, when `_UseNormalMap` enabled); 0 (Meta SDK, normal decode handled by SDK pipeline)

V3 retains the V2 toon quantisation as the surface shading base; the detected edge mask E is composited over it by linear interpolation toward the configurable line colour `_InnerLineColor`. The three exposed shader parameters are `_NormalEdgeThreshold` (τ, smoothstep centre), `_NormalEdgeSmoothness` (σ, half-width), and `_NormalEdgeStrength` (α_N, edge weight scale).

Meta SDK. On the Meta SDK platform, the world-space normal supplied to `ddx/ddy` is `i.geometry.normal` from `AppSpecificPostManipulation` — the SDK's fully composited fragment normal, which already incorporates the SDK's internal normal-map decode. No additional texture read is required:

```
float3 dNdx = ddx((float3)worldNormal);
float3 dNdy = ddy((float3)worldNormal);
float  M    = sqrt(dot(dNdx, dNdx) + dot(dNdy, dNdy));

float nMin = _NormalEdgeThreshold - _NormalEdgeSmoothness;
float nMax = _NormalEdgeThreshold + _NormalEdgeSmoothness;
float edge = smoothstep(nMin, nMax, M) * _NormalEdgeStrength;
```

Listing 5.8: V3 normal gradient and edge weight : Meta SDK (`NPREffect_NormalEdge.cginc`).

Mixamo and Avaturn. The normal must be decoded from the PBR normal map before the derivative is computed. The vertex shader builds the tangent frame from the mesh tangent attribute; the fragment shader then unpacks the normal map and transforms it from tangent space to world space via the TBN matrix:

```
// vertex shader output (Varyings)
float3 tangentWS   : TEXCOORD4;
float3 bitangentWS : TEXCOORD5;

// fragment shader -- decode normal map then apply derivatives
float3 geomNWS = normalize(IN.nWS);
float3 nWS     = geomNWS;

if (_UseNormalMap > 0.5)
{
    float3 normalTS = UnpackNormalScale(
        SAMPLE_TEXTURE2D(_BumpMap, sampler_BumpMap, IN.uv),
    _BumpScale);
    float3x3 TBN = float3x3(
        normalize(IN.tangentWS),
        normalize(IN.bitangentWS),
        geomNWS);
    nWS = normalize(mul(normalTS, TBN));
}

// nWS is then supplied to ddx/ddy as above
```

Listing 5.9: V3 normal map decode via TBN : Mixamo and Avaturn (`V2_NormalEdgeDetection.shader`).

When `_UseNormalMap` is disabled, `nWS` falls back to the bare interpolated geometry normal. The normal map files used are `Ch37_1001_Normal` and `Ch37_1002_Normal` for Jody, and the glTF-exported normal map from the Avaturn pipeline.

The distance-dependent edge thickening described in Section 4.3.5 motivated the UV-space albedo sampling approach of V4.

Table 5.3: V3 implementation comparison across the three avatar platforms.

	Jody (Mixamo)	**Avaturn**	**Meta SDK**
Normal source	TBN-decoded PBR normal map (baked)	TBN-decoded PBR normal map (glTF)	SDK composited fragment normal
Edge detail	Medium — normal map + mesh curvature	Dense — fine microdetail in normal map	Medium — SDK mesh anatomy
Sample cost	1 (normal map); 0 when disabled	1 (normal map); 0 when disabled	0 extra samples
Integration point	URP ForwardLit pass	URP ForwardLit pass	`AppSpecificPostManipulation`

Visual results for V3 across all three platforms, including the distance-dependent edge thickening, are documented in Figure 4.10 in Chapter 4.

5.4.4 V4 — Sobel Edge Detection

Shader: `Assets/Shaders/MetaAvatarShaders/NPREffect_Sobel.cginc` (Meta SDK); URP forward-lit pass with albedo sampling (Mixamo, Avaturn)
Extra texture samples per fragment: 9 (one per 3×3 neighbourhood position)

V4 moves from geometry-derived signals to the diffuse albedo texture as the edge source. Each of the nine 3×3 neighbourhood positions is sampled and converted to a scalar luminance using the ITU-R BT.601 luma coefficients [11]:

$$L = 0.299\,R + 0.587\,G + 0.114\,B \tag{5.1}$$

This prevents colour channel cross-talk from introducing false edges at hue boundaries where RGB channels change in opposite directions but luminance is constant. The Sobel kernels K_x and K_y are then applied to the nine luminance samples (kernels defined in Equation 3.2 in the background chapter).

```
float3 L      = float3(0.299, 0.587, 0.114);
float  offset = _SobelSampleDist * 0.001;

float tl = dot(tex2D(u_BaseColorSampler, uv+float2(-offset, offset)
   ).rgb, L);
```

```
float t  = dot(tex2D(u_BaseColorSampler, uv+float2(      0, offset)).rgb, L);
float tr = dot(tex2D(u_BaseColorSampler, uv+float2( offset, offset)).rgb, L);
float l  = dot(tex2D(u_BaseColorSampler, uv+float2(-offset,      0)).rgb, L);
float r  = dot(tex2D(u_BaseColorSampler, uv+float2( offset,      0)).rgb, L);
float bl = dot(tex2D(u_BaseColorSampler, uv+float2(-offset,-offset)).rgb, L);
float b  = dot(tex2D(u_BaseColorSampler, uv+float2(      0,-offset)).rgb, L);
float br = dot(tex2D(u_BaseColorSampler, uv+float2( offset,-offset)).rgb, L);

float Gx      = (tr + 2*r + br) - (tl + 2*l + bl);
float Gy      = (tl + 2*t + tr) - (bl + 2*b + br);
float edgeMag = sqrt(Gx*Gx + Gy*Gy);
```

Listing 5.10: V4 nine-tap luminance sampling and Sobel gradient (`NPREffect_Sobel.cginc`).

UV seam boundaries produce spurious detections: texels on either side of a seam carry distant UV coordinates, so the colour difference registers as a strong edge even though no visual boundary exists. The suppression computes the luminance spread across all nine samples and rejects detections where the spread exceeds `_SobelSeamLimit`. A second upper bound `_SobelMax` clips spike magnitudes that exceed any plausible structural edge:

```
float lMin = min(min(min(tl,t),min(tr,l)), min(min(r,bl),min(b,br)));
float lMax = max(max(max(tl,t),max(tr,l)), max(max(r,bl),max(b,br)));

float seamMask = step(lMax - lMin, _SobelSeamLimit);   // reject seam spikes
float inBand   = step(_SobelThreshold, edgeMag)
               * step(edgeMag, _SobelMax);             // accept only valid range

float edge = inBand * seamMask * _SobelStrength;
color.rgb  = lerp(color.rgb, _InnerLineColor.rgb, edge);
```

Listing 5.11: Seam suppression and magnitude band gate.

The UV-space offset `_SobelSampleDist` is distance-independent, which resolves the distance artifact of V3: the sampling neighbourhood stays fixed in texture space regardless of how far the avatar is from the camera.

On Mixamo and Avaturn, `u_BaseColorSampler` reads the raw diffuse albedo, so edge detection responds to surface material boundaries. On the Meta SDK, the hook receives only the composited PBR output; the Sobel therefore responds to shadow boundaries and

specular gradients in addition to material transitions, which reduces selectivity but still adds visible clothing-level edges and broad facial contour lines.

Table 5.4: V4 implementation comparison across the three avatar platforms.

	Jody (Mixamo)	**Avaturn**	**Meta SDK**
Sobel input	Raw diffuse albedo texture	Raw diffuse albedo texture	Composited PBR colour
Luminance weights	BT.601 (0.299 / 0.587 / 0.114)	BT.601	BT.601
Sample count	9 albedo samples	9 albedo samples	9 samples on PBR output
Seam suppression	Luma range gate + upper magnitude bound	same	same
Distance artifact	None (UV-space offset)	None	None
Shadow edges	No (pre-lighting albedo)	No	Yes (PBR composite)

Visual results for V4 across all three platforms are documented in Figure 4.11 in Chapter 4.

5.4.5 V4.2 — Gaussian Prefiltered Sobel

Shaders: `Assets/Shaders/MetaAvatarShaders/NPREffect_GaussianSobel.cginc` (Meta SDK); `Assets/Shaders/JadeNPRShaders/V4_GaussianPreFilteredSobel.shader` (Mixamo); `Assets/Shaders/AvaturnNPRShaders/V4_GaussianPreFilteredSobel.shader` (Avaturn)
Keyword (Meta): `EFFECT_GAUSS_SOBEL`
Extra texture samples per fragment: up to 72 (8 Sobel positions × 9 Gaussian taps) when blur is enabled; 8 when disabled

The pre-filter is implemented as a 9-tap weighted luminance helper (`GaussianLuma` on the Meta SDK, `SampleLuminanceBlurred` on Mixamo and Avaturn). At each call, one centre sample, four axis-aligned cardinal samples, and four diagonal corner samples are drawn at a configurable offset from the given UV position. The three weight groups are normalised to unity before use so that the output is always a valid luminance estimate:

```
1 float GaussianLuma(float2 center, float blurR, float cW, float
     cardW, float diagW)
2 {
3     float3 L = float3(0.299, 0.587, 0.114);
4     float  v = 0.0;
5     v += dot(tex2D(u_BaseColorSampler, center).rgb, L) * cW;
6     v += dot(tex2D(u_BaseColorSampler, center + float2( blurR,
     0)).rgb, L) * cardW;
7     v += dot(tex2D(u_BaseColorSampler, center + float2(-blurR,
     0)).rgb, L) * cardW;
```

```
    v += dot(tex2D(u_BaseColorSampler, center + float2(     0,
    blurR)).rgb, L) * cardW;
    v += dot(tex2D(u_BaseColorSampler, center + float2(     0, -
    blurR)).rgb, L) * cardW;
    v += dot(tex2D(u_BaseColorSampler, center + float2( blurR,
    blurR)).rgb, L) * diagW;
    v += dot(tex2D(u_BaseColorSampler, center + float2(-blurR,
    blurR)).rgb, L) * diagW;
    v += dot(tex2D(u_BaseColorSampler, center + float2( blurR, -
    blurR)).rgb, L) * diagW;
    v += dot(tex2D(u_BaseColorSampler, center + float2(-blurR, -
    blurR)).rgb, L) * diagW;
    return v;
}

// Weight normalisation: computed once before the nine Sobel
    positions are sampled
float totalW = _GSobelCenterWeight + 4.0*_GSobelCardinalWeight +
    4.0*_GSobelDiagonalWeight;
totalW = max(totalW, 0.0001);
cW    = _GSobelCenterWeight   / totalW;
cardW = _GSobelCardinalWeight / totalW;
diagW = _GSobelDiagonalWeight / totalW;
```

Listing 5.12: 9-tap Gaussian luminance helper with weight normalisation (`NPREffect_GaussianSobel.cginc`).

When `_GSobelEnableGaussBlur = 0` the weights collapse to centre $= 1.0$, cardinal $= 0$, diagonal $= 0$, making the helper a plain point sample that reproduces V3 behaviour at no additional cost. The helper is called once at each of the nine Sobel neighbourhood positions, and the G_x / G_y kernels are applied to the nine resulting luminance values using the same formula as V3. After computing the gradient magnitude, a configurable threshold band gates the result: `smoothstep(_GSobelThreshold * _GSobelThreshMin, _GSobelThreshold * _GSobelThre` `edgeMag)`. The upper ceiling of this band suppresses extreme-magnitude detections such as UV seam spikes, replacing the explicit luma-range seam gate of V3.

A series of four progressive smoothstep passes then sharpens the thresholded magnitude into a crisp edge mask:

```
float hw1 = lerp(0.5, 0.03, _GSobelTightness);
edge = smoothstep(0.5 - hw1, 0.5 + hw1, edge);
float hw2 = lerp(0.5, 0.15, _GSobelTightness);
edge = smoothstep(0.5 - hw2, 0.5 + hw2, edge);
float hw3 = lerp(0.5, 0.25, _GSobelTightness);
edge = smoothstep(0.5 - hw3, 0.5 + hw3, edge);
float hw4 = lerp(0.5, 0.35, _GSobelTightness);
edge = smoothstep(0.5 - hw4, 0.5 + hw4, edge);
edge = pow(edge, _GSobelPowerCurve);
edge *= _GSobelStrength;
```

Listing 5.13: 4-pass progressive smoothstep sharpening and power curve (`NPREffect_GaussianSobel.cginc`).

At `_GSobelTightness` = 0 every half-width equals 0.5, reducing each pass to a linear identity; at `_GSobelTightness` = 1 the widths shrink to 0.03, 0.15, 0.25, and 0.35, composing a steep sigmoid. `_GSobelPowerCurve` > 1 suppresses soft halos; values below 1 amplify thin lines.

Platform differences. On the Meta SDK, all four smoothstep passes are always evaluated and driven by a single `_GSobelTightness` scalar. The Mixamo and Avaturn shaders (`V4_GaussianPreFilteredSobel.shader`) expose the four passes as individually toggleable flags (`_EnablePass1`–`_EnablePass4`) and offer three preset configurations for rapid cross-avatar setup: Light filtering uses the classical kernel weights (centre 0.25, cardinal 0.125, diagonal 0.0625, blur multiplier 0.8, power curve 1.5); Moderate uses a wider footprint (blur multiplier 1.0, centre 0.3, power curve 2.0); Aggressive applies a narrow threshold band (0.85–1.15), maximum tightness, and a steep power curve of 3.0. The default parameter set on all three platforms corresponds to the Light configuration.

Table 5.5: V4.2 implementation comparison across the three avatar platforms.

	Jody (Mixamo)	**Avaturn**	**Meta SDK**
Gaussian input	Raw diffuse albedo	Raw diffuse albedo (glTF)	Composited PBR output
Default kernel weights	centre 0.25 / cardinal 0.125 / diagonal 0.0625	same	same
Seam suppression	Threshold band (min/max multipliers)	same	same
Smoothstep passes	4, individually toggleable (`_EnablePass1`–`_EnablePass4`)	same	4, always on via `_GSobelTightness`
Preset configurations	Light / Moderate / Aggressive	same	N/A
Sample count (blur on)	Up to 72	Up to 72	Up to 72

Visual results for V4.2 across all three platforms are documented in Figure 4.13 in Chapter 4.

5.4.6 V5 — Hierarchical Edge Detection

The hierarchical implementation computes three independent edge layers in a single fragment shader invocation and combines them via weighted addition.

The depth layer uses `ddx` and `ddy` applied to the fragment's linear eye depth to estimate the depth gradient magnitude. The gradient is normalised by `_DepthSensitivity` and thresholded to produce a binary edge signal. The normal layer applies the same derivative approach to the interpolated world-space surface normal vector, computing $M_N = \sqrt{\|\texttt{ddx}(\mathbf{N})\|^2 + \|\texttt{ddy}(\mathbf{N})\|^2}$ and thresholding against `_NormalThreshold`. The colour layer applies the Roberts Cross operator to the diffuse texture: it samples four neighbours in a 2×2 diagonal pattern and computes the cross-difference magnitudes $|R_1|$ and $|R_2|$ as defined in Section 3.2.1, thresholded against `_ColorThreshold`.

The three binary edge signals are weighted by `_DepthWeight`, `_NormalWeight`, and `_ColorWeight` respectively and summed. The combined weight is clamped to $[0, 1]$ and used to blend between the base albedo and `_EdgeColor`.

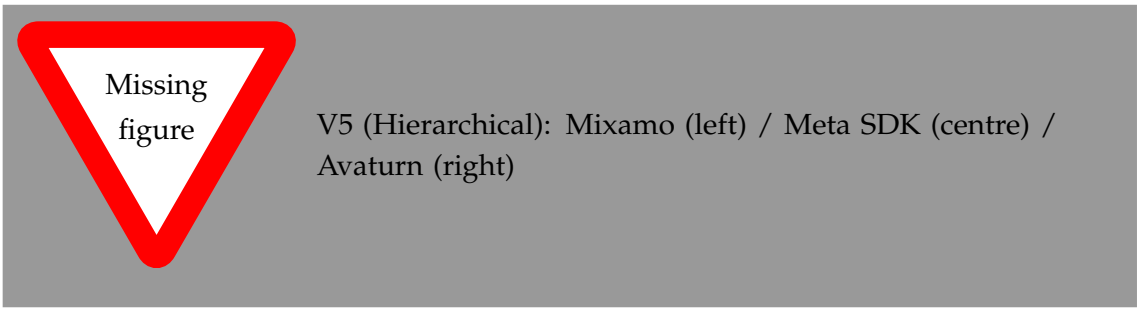


Figure 5.1: V5 (Hierarchical Edge Detection) across the three avatar platforms.

5.4.7 V6 — Kuwahara Painterly Filter

The Kuwahara implementation evaluates four overlapping quadrant subregions around the current fragment. Each quadrant is a square of radius `_KuwaharaRadius`, offset from the centre so that adjacent quadrants share a one-texel border. For each quadrant, the implementation accumulates the sum and sum-of-squares of the sampled colour values, then computes the per-channel mean and variance from these accumulators. The quadrant with the lowest luminance variance is selected, and its mean colour is written as the fragment output.

Because the filter operates on the diffuse texture using fixed texture-coordinate offsets, no intermediate render target is required, and the entire computation runs inside the single-pass fragment hook. The isotropic formulation is used; the anisotropic extension of Kyprianidis et al. [13] requires a structure tensor computed in a prior pass and is therefore incompatible with the single-pass constraint of the Meta SDK hook.

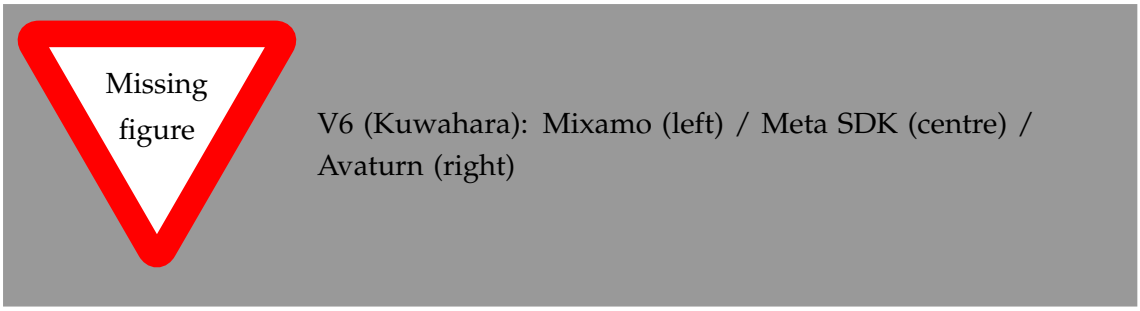


Figure 5.2: V6 (Kuwahara Painterly Filter) across the three avatar platforms.

5.4.8 V7 — Combination: Kuwahara, Gaussian Sobel, and Hierarchical

The V7 implementation chains three stages within the same fragment shader invocation. The Kuwahara filter is applied first, producing a smoothed mean colour from the lowest-variance quadrant. The Gaussian prefiltered Sobel gradient is then computed on the Kuwahara output rather than the raw diffuse texture; because the Kuwahara output is piecewise uniform within regions, the Sobel operator responds primarily to inter-region boundaries, suppressing the fine texture gradients that produce noise in V3. The seam rejection checks from V3 are retained. Finally, the hierarchical edge layers from V5 — depth proximity, normal discontinuity, and Roberts Cross colour — are composited on top with independent per-layer weights. On the Meta SDK platform, the Sobel stage is omitted; both the Sobel operator and the Roberts Cross component of the hierarchical layer detect colour-based boundaries and would double-count the same edge signal within the single-pass hook (see Section 5.4.8).

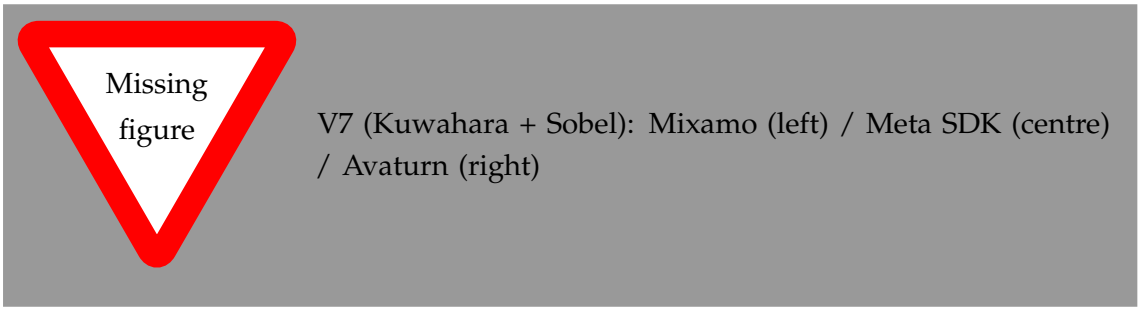


Figure 5.3: V7 (Combination: Kuwahara, Gaussian Sobel, and Hierarchical) across the three avatar platforms. The Meta SDK variant omits the Sobel stage.

5.4.9 V8 — Custom

V8 is a custom shader developed specifically for the Avaturn avatar's head material, reflecting the decision to proceed with Avaturn as the primary platform for the user study. Rather than applying a uniform stylisation pipeline across the full mesh, this variant assigns different per-region parameter sets to the head and clothing submeshes. The pipeline follows the same three-stage structure as V7 — Kuwahara base, Gaussian prefiltered Sobel edges, hierarchical multi-cue overlay — but the Sobel threshold is elevated on the head material to suppress the baked-illumination artefact documented in

V4 and V4.2 (see Section 4.3.7), while the clothing material retains a lower threshold to preserve fabric detail. The per-material parameter separation is managed through the `AvaturnSlotPresets` editor; all other parameters remain identical across submeshes.

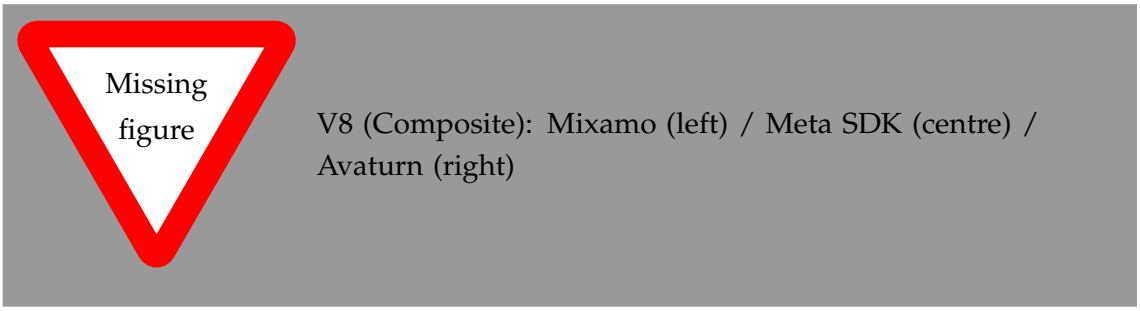


Figure 5.4: V8 (Custom: Avaturn head-specific shader) applied to the Avaturn avatar, showing per-region parameter separation between head and clothing.

An additional extension to the V8 pipeline introduced chromatic edge detection targeting the lip region. Rather than applying the Sobel operator to luminance across the full texture, this technique first computes a colour-proximity mask by comparing each texel against a reference lip colour using a per-fragment distance metric gated by a tolerance parameter L_{tol} (`_LipColorTolerance`). The Sobel gradient is then applied to this mask rather than to raw luminance, so edge responses fire at the boundary between lip-coloured pixels and the surrounding skin tone rather than at every texture contrast transition. The edge mask is further gated by a threshold L_t (`_LipEdgeThreshold`), which controls the minimum gradient magnitude required to register as a lip boundary line.

Figure 5.5 shows the effect of varying L_{tol} from 0.05 to 0.15 while keeping L_t fixed at 0.155. At low tolerance values only pixels closely matching the reference lip colour are included in the mask, producing a narrow and fragmented detection region; increasing the tolerance progressively admits adjacent skin-tone pixels, broadening the mask and producing a more contiguous but also over-expanded region that begins to include the surrounding face.

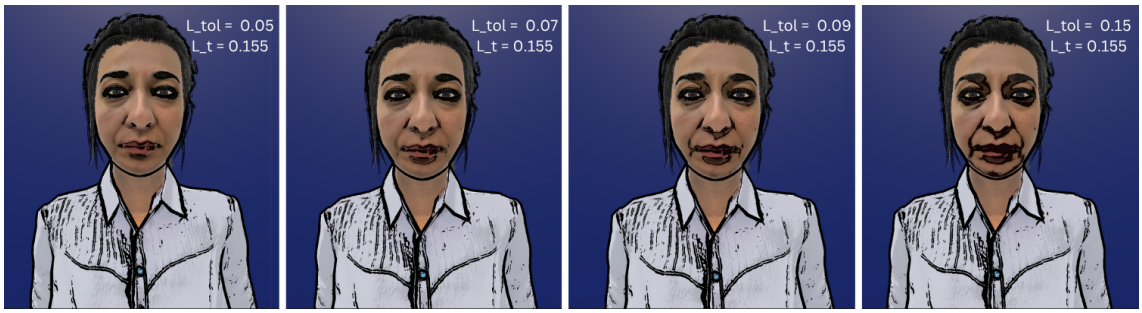


Figure 5.5: Effect of varying lip colour tolerance L_{tol} (0.05, 0.07, 0.09, 0.15) with lip edge threshold L_t held constant at 0.155. Low tolerance restricts the colour-proximity mask to close lip-colour matches; higher values expand the mask into adjacent skin tones.

Figure 5.6 shows the effect of varying L_t from 0.1 to 0.5 with L_{tol} held constant at 0.1. Low threshold values accept weak gradient responses across the full mask boundary,

yielding dense lines but including spurious detections from mask noise; raising the threshold suppresses weaker responses and produces a cleaner, thinner outline, at the cost of missing lower-contrast boundary segments.

Figure 5.6: Effect of varying lip edge threshold L_t (0.1, 0.2, 0.3, 0.4, 0.5) with L_{tol} held constant at 0.1. A low threshold accepts weak gradient responses and produces a dense, noisy line; a high threshold retains only strong responses, yielding a cleaner outline but potentially missing faint boundary segments.

The technique did not produce reliable results in practice. The Avaturn head texture contains baked ambient occlusion and subtle illumination gradients across the skin surface, so the colour content is not spatially uniform. As a consequence, the colour-proximity mask produced inconsistent matches, firing in shadow regions near the nose bridge and chin that shared the same colour range as the lip target. Adjusting L_{tol} could not simultaneously suppress these false matches and retain complete lip coverage. The approach was therefore set aside. The head texture was instead modified in Blender: baked ambient occlusion and illumination gradients were reduced to produce a more uniform skin tone across the face, without altering the avatar's geometry, proportions, or facial identity. With this revised texture, the core V8 pipeline was reapplied to the head material, where the reduced illumination variation allowed the Sobel thresholds to be tuned more precisely without interference from false chromatic matches. The original unmodified texture is retained for the photorealistic baseline condition (C2) of the user study; the modified texture is used exclusively for the NPR condition (C3). The implications of this distinction for the C2–C3 comparison are addressed in Section 6.1.4.

Following the outcome of the pre-study ranking exercise (Section 4.6), the Kuwahara and hierarchical edge detection materials are not assigned to either the head or clothing submesh in the final C3 configuration; only the V4 Gaussian pre-filtered Sobel material is applied, retaining the per-region threshold separation between head and clothing established earlier in this section. Table 5.6 lists the final property values applied to this material for the C3 stimulus videos.

Table 5.6: Final V4 Gaussian pre-filtered Sobel shader property values applied to the C3 stimulus, as determined by the pre-study ranking exercise (Section 4.6).

Shader property	Value
`_InnerLineThreshold`	0.03
`_OuterOutlineWidth`	0.005
`_UseOutlineDepthOffset`	On
`_ShadowStrength`	0.5
`_DetailMode`	Depth
`_DetailBias`	0.5
`_EnableRim`	On (`_RimPower` = 4.0)

5.5 Runtime Tooling

Two runtime utilities were developed to address constraints that cannot be resolved at shader compilation time: the first enables interactive parameter exploration within the VR headset during the visual assessment phase; the second manages material assignment on the Meta SDK platform, where the SDK's LOD system reverts custom shader assignments automatically.

5.5.1 In-VR Parameter Tuning Interface

Shader parameters can be adjusted through two routes: the Unity Inspector during offline development, and a custom in-VR WorldSpace panel during live headset sessions. The Inspector provides the standard Unity material property interface, allowing each parameter to be set numerically or through drag controls on the material asset before building. The WorldSpace panel, implemented in `NPREdgeDetectionUI.cs`, provides the same controls in the running application so that parameter values can be explored and compared while the avatar is visible in the headset at correct VR scale.

The panel is constructed at runtime from Unity UI primitives: each tunable parameter is represented by a labelled row containing either a slider or a toggle button, laid out in technique-specific groups. Controller raycasting using the OVR interaction layer detects when the ray from either controller intersects a slider's `BoxCollider`; a trigger-pull gesture initiates a drag interaction that maps horizontal ray movement to the slider's value range. Technique switching is handled through a dropdown row: selecting a technique calls `EnableKeyword` and `DisableKeyword` on the active materials to swap the active shader branch without recompilation. The panel exposes parameters across all technique groups, including per-technique thresholds, weights, kernel radii, and colour properties, and was central to the visual assessment process used to select the technique for the user study.

5.5.2 LOD Material Management

`AvatarShaderSwapper.cs` solves the material reversion problem described in Section 5.2.2. On startup it subscribes to the avatar load completion callback and performs an initial material replacement pass across all LOD levels. A coroutine running at a configurable

polling interval inspects the active materials on each renderer at runtime; if an SDK-assigned material is detected — indicating that a LOD transition has reverted the custom assignment — the coroutine re-applies the NPR material and copies the base colour and normal texture references from the newly active SDK material into the replacement shader's properties.

5.5.3 Pose Freeze for Evaluation

`AvatarFreezeController.cs` enables the avatar's body-tracking pose to be frozen at a chosen instant, preserving a specific expression and body posture for evaluation screenshots. The freeze is implemented by accessing the `_inputTrackingProvider` field of `OvrAvatarInputManager` via reflection and substituting a snapshot provider that returns a fixed pose state. Pressing A, X, or the Space key toggles the freeze on and off.

6 Evaluation

This chapter evaluates the NPR avatar system developed in this thesis. The first part documents the visual quality assessment of the eight NPR techniques across the three avatar platforms, establishing which technique was selected for the user study and the reasoning behind that selection. The second part presents the user study that tests whether NPR stylisation affects how participants perceive trust in a speaking avatar.

6.1 Visual Quality Assessment

6.1.1 Assessment Methodology

Each NPR technique was evaluated qualitatively through a structured visual assessment conducted inside the Unity Editor and on the Meta Quest headset using the in-VR parameter panel described in Section 5.5.1. The assessment followed a consistent procedure: the avatar was placed in the shared studio-lighting environment described in Section 5.2.4, animated with a repeating idle loop, and observed at the conversational framing distance of approximately 1 m from the camera — the distance used to record the user study videos. Parameters were adjusted interactively through the WorldSpace panel until the visual output satisfied four criteria:

1. **Facial readability.** Facial features — eyes, mouth boundary, nose bridge, and jaw line — must remain individually legible under stylisation. A technique that obscures or merges these features cannot serve as a plausible avatar representation in the study.
2. **Temporal stability.** Edges and stylisation effects must remain stable during animation. Screen-space techniques are particularly susceptible to flickering when the avatar translates across the screen, because the pixel-space neighbourhood shifts while the underlying geometry does not.
3. **Artefact suppression.** Visible UV seam lines, rectangular outlines around hair cards, depth-fighting at the outline hull, and false edge detections at skin-tone boundaries must be suppressed to a level that is not noticeable under normal observation.
4. **Expressive abstraction.** The stylisation should be visually legible as an intentional artistic choice, not as a rendering error. A technique that produces only isolated noise marks rather than coherent contours would not read as deliberate stylisation.

Evaluation proceeded iteratively for each technique. Parameters were first tuned on the Mixamo avatar, where unrestricted shader access made per-pass debugging straightforward, and then verified on Avaturn and the Meta SDK avatar. Techniques that satisfied all four criteria on at least two of the three platforms were considered viable candidates for the user study.

6.1.2 Per-Technique Assessment

V1 — Toon Shading. Toon shading satisfies facial readability and temporal stability because the quantised diffuse term produces a consistent, low-frequency tonal structure that does not flicker. Four toon bands provide sufficient depth cues to read facial shape at the 1 m framing. The limitation is absence of interior line work: the inverted hull outline marks the external silhouette only, and no edges are drawn at material boundaries or creases. Clothing seams and facial features that rely on local contrast rather than silhouette curvature are not delineated. V1 is included in the study materials as the simplest abstraction reference.

V1.2 — X-Toon Extended Toon Shading. The X-Toon technique replaces procedural band quantisation with a 2D ramp texture, introducing an abstraction level axis that progressively flattens shading as a function of camera distance or surface curvature. At the 1 m conversational framing used in the study, the depth range across the avatar's torso and face spans only a few centimetres, which is too narrow relative to the configured depth window to produce meaningful spatial variation in the abstraction level. Curvature mode produced near-zero screen-space normal derivatives over the smooth head and body geometry, making the V axis equally inert. As a result, the visual output of V1.2 at the study framing distance is perceptually indistinguishable from V1. The technique was retained as a technically motivated component — the 2D ramp provides superior control over tonal response compared to procedural quantisation — but was not selected for the study because its signature feature is not perceptible under study conditions.

V2 — Screen-Space Normal Edge Detection. V2 catches silhouette edges and creases through the `ddx/ddy` derivative applied to the world-space surface normal, adding interior contour detail that V1 lacks. The known limitation is the distance dependence of pixel-space derivatives: a fixed-world-space crease subtends fewer pixels as camera distance increases, causing the derived gradient magnitude to rise and edges to thicken with distance rather than thin. At the 1 m framing, this effect is mild and edges are perceptually stable. V2 has zero additional texture samples, making it the least costly technique after V1. The edges it detects are limited to geometry and normal-map discontinuities; colour boundaries at material seams are not detected.

V3 — Sobel Edge Detection. V3 detects colour boundaries in the diffuse albedo texture by applying the Sobel operator to a 3×3 luminance neighbourhood. This adds material-boundary lines — the edge between skin and clothing, the contour of eyebrows, the lip boundary — that V2 cannot detect. The identified problem is UV seam detection: texels on opposite sides of a UV seam carry distant coordinate values, so the colour difference registers as a strong edge across the seam. This artefact was addressed with a luminance range gate (`_SeamRangeLimit`) and an upper magnitude bound (`_SobelMax`). After tuning, seam lines are suppressed to below the detection threshold on both the Mixamo and Avaturn avatars under the studio lighting configuration. The skin-to-clothing boundary produces the most visually significant lines on the Avaturn avatar; eyebrow and lip boundaries emerge at moderate threshold settings.

V3.2 — Gaussian Prefiltered Sobel. V3.2 wraps each of the nine Sobel neighbourhood samples in a 9-tap Gaussian helper, producing a pre-smoothed luminance signal before gradient computation. Smoothing reduces noise in the gradient output, producing cleaner, more coherent edge lines at the cost of up to 72 texture samples per fragment. Perceptually, the improvement over V3 is visible as a reduction in thin noise marks on clothing with high-frequency texture; structural lines at material boundaries are unaffected. The seam suppression from V3 is retained. At the 1 m framing, the visual difference between V3 and V3.2 is modest, suggesting that the noise reduction benefit is not sufficient to justify the sample count increase for a study condition where only a single technique is presented.

V5 — Hierarchical Multi-Cue Edge Detection. V5 combines three independent edge signals — a depth gradient, a normal gradient, and a Roberts Cross colour edge — with per-layer weights. The combination addresses the selectivity limitation of each individual cue: depth edges mark silhouette breaks caused by proximity differences; normal edges mark geometric creases; Roberts Cross marks colour boundaries using only four samples per fragment rather than nine. The result is a technique that captures a wider range of structural boundaries than any single cue, producing strong contour lines at outer silhouettes, contour lines at garment folds, and faint markings at skin-tone transitions, simultaneously. Temporal stability is comparable to V2 because the depth and normal layers are computed from the same pixel-space derivatives, while the Roberts Cross colour layer adds four albedo samples. Artefact suppression follows from the colour layer's shorter neighbourhood: the 2×2 Roberts Cross footprint is less susceptible to UV seam detection than the 3×3 Sobel neighbourhood.

V6 — Kuwahara Painterly Filter. V6 produces a painterly abstraction of the diffuse texture by replacing each pixel's colour with the mean of the lowest-variance quadrant neighbourhood. The visual effect is a piecewise-uniform fill with sharp boundaries between colour regions, approximating the look of a painted surface. The limitation is identity loss: at radius values large enough to produce a clearly painterly effect, fine features including the eyebrows, lip boundary, and iris colour are flattened or merged with the surrounding skin tone. Reducing the radius preserves features but reduces the painterly effect to the point where the output is visually similar to a blurred or smoothed albedo rather than a deliberate style. V6 is the only technique that modifies the colour fill rather than adding edge marks; it was therefore identified as potentially effective in combination with edge-based techniques, rather than as a standalone study condition.

V7 and V8 — Composite Techniques. V7 chains the Kuwahara fill (V6) with Gaussian Sobel edge detection (V3.2) and the hierarchical overlay (V5) in a single fragment shader invocation. The Kuwahara fill provides painterly surface abstraction; the Gaussian Sobel edges overlay ink-line contours at material boundaries on the unified colour output; the hierarchical layer adds depth and normal edges at silhouettes and creases. The three stages reinforce each other: the Kuwahara output is piecewise uniform, so Sobel applied to it fires primarily at structural boundaries rather than at texture noise. Temporal stability is the main risk in V7 because the Sobel stage operates on the Kuwahara output, and the Kuwahara stage is itself computed from multiple texture samples at a fixed UV offset. Under head motion, the sampling neighbourhood translates with the avatar, producing

subtle flicker at region boundaries. This was mitigated by widening the smoothstep threshold and reducing the Sobel strength parameter relative to their standalone values.

V8 is the variant of V7 adapted specifically for the Avaturn avatar's head material. The head material carries baked ambient occlusion and subtle illumination gradients that produce false Sobel detections across the face in the standard V7 configuration. V8 addresses this by elevating the Sobel threshold on the head material while preserving a lower threshold on the clothing materials, where fabric texture provides legitimate Sobel edges. The chromatic lip-edge extension developed for V8 was discontinued after it produced inconsistent results on the Avaturn head texture (see Section 5.4.9).

6.1.3 Technique Comparison Summary

Table 6.1: Summary of visual quality assessment results across the eight NPR techniques on the Avaturn platform. Criteria are rated qualitatively: ✓ = satisfactory without tuning, (✓) = satisfactory after tuning, × = limitation observed.

Technique	Facial readability	Temporal stability	Artefact suppression	Expressive abstraction
V1 — Toon	✓	✓	✓	(✓) no interior lines
V1.2 — X-Toon	✓	✓	✓	× V-axis inert at 1 m
V2 — Normal edge	✓	✓	✓	(✓) no colour edges
V3 — Sobel	✓	(✓)	(✓) seam gate	✓
V3.2 — Gaussian Sobel	✓	(✓)	(✓) seam gate	✓
V5 — Hierarchical	✓	✓	✓	✓
V6 — Kuwahara	× at large r	✓	✓	✓
V7/V8 — Composite	✓	(✓)	(✓) head param	✓

6.1.4 Technique Selection for the User Study

The Sobel, outline, and XToon stages of the V8 configuration were selected for the user study condition C3, retaining the per-region head and clothing threshold separation introduced in V8. The selection was based on two considerations. First, elevating the Sobel threshold on the head material suppresses the UV seam and baked-illumination artefacts identified in Section 4.3.7 while facial features remain individually legible. Second, the outline, Sobel, and XToon combination reads clearly as an intentional stylisation rather than a rendering artefact, which is necessary for the manipulation check embedded in the study questionnaire (Part I); a technique that produces only subtle modifications to the photorealistic baseline might fail the manipulation check, reducing usable data. The Kuwahara painterly fill and the hierarchical depth edge layer were both evaluated as candidate additions to this composite but were excluded following the pre-study ranking exercise, as detailed in Section 4.6.

The Avaturn photorealistic baseline (condition C2) uses the same avatar mesh, rig, animation, and scene configuration as C3, with only the material assignment changed. This

ensures that any measured difference in trust or eeriness between C2 and C3 is attributable to the rendering style rather than to differences in geometry, animation quality, or scene context.

One technical exception applies to the albedo texture of the head submesh. The C3 condition uses a version of the head texture modified in Blender to reduce baked ambient occlusion and illumination gradients across the skin surface. This preprocessing was required because the Gaussian pre-filter alone could not suppress the false Sobel detections that the baked-shading variation produced across the face. The C2 condition retains the original unmodified Avaturn texture. The modification does not alter facial geometry, proportions, or perceived identity; it reduces low-level colour variance in the skin region without changing any structural or expressive feature of the avatar. Because the C3 pipeline applies the Sobel operator to structural boundaries rather than fine texture gradients, the modification is not perceptually distinguishable in the C3 rendered output. The C2 baseline continues to represent the genuine Avaturn avatar without shader or texture modification.

With the C3 pipeline and texture configuration established, a parameter selection pilot study was conducted to determine the specific shader parameter values used in C3, and to test whether the Kuwahara and hierarchical post-processing passes should be included in the composite at all. The design of this pre-study, its statistical analysis, and the outcome for each parameter group are reported in Section 4.6. Both post-processing passes were excluded from the final configuration: participants ranked the unmodified baseline as significantly more trustworthy than every tested variant of each pass. Parameters with comparatively minor expected visual impact, including rim light presence and Kuwahara kernel size, were set through direct visual inspection during development rather than included in the ranking exercise, to keep the pilot study within a manageable length for participants.

6.2 User Study Results

The user study was designed and conducted following the protocol described in Section 4.7. This section reports the findings. Full descriptive statistics, raw scale scores, and counterbalancing verification tables are provided in Appendix A.

6.2.1 Participants

Table 6.2: Participant demographics and Latin Square group distribution.

Parameter	**Value**
Total enrolled	*[N]*
Excluded (comprehension check)	*[n]*
Final analytic sample	*[N_{final}]*
Age: mean (SD)	*[M] ([SD])* years
Gender	*[breakdown]*
Latin Square groups (G1–G6)	*[n per group]*

Participants were recruited via [recruitment channel]. Inclusion criteria required age 18 or older, sufficient proficiency in the study language to understand the vignette scenarios, and access to the study URL. No specific domain expertise was required. Participants who selected the wrong option on the comprehension check (Part A) for any block were excluded from the analytic sample.

6.2.2 Manipulation Check

Part I of the questionnaire asked participants to rate how visually stylised each advisor appeared on a single seven-point scale ranging from completely realistic (1) to very stylised (7), administered once after all three video blocks. This single-item manipulation check was designed to verify that participants perceived the rendering difference between conditions without priming stylisation awareness during the rating blocks. A successful manipulation requires that C3 (NPR) scores significantly higher than C2 (photorealistic), while C1 (real person) scores lowest.

[Insert Part I descriptive statistics and one-way repeated-measures ANOVA results here. Expected pattern: C1 < C2 < C3 on the stylisation scale. Report F, df, p, η_p^2 and pairwise comparisons with Bonferroni correction. If the C2 vs C3 comparison is not significant, the manipulation has failed and trust results must be interpreted with caution.]

6.2.3 Hypothesis Tests

The primary analysis was a one-way repeated-measures MANOVA with condition (C1, C2, C3) as the within-subjects factor and five subscale composite scores (B_score, D_score, E_score, F_score, G_score) as dependent variables. Significant omnibus effects were followed up with univariate repeated-measures ANOVAs per subscale and pairwise comparisons with Bonferroni correction ($\alpha_{\text{corrected}} = .05/3 = .017$). Effect sizes are reported as partial η^2.

H1 — NPR Reduces Eeriness

H1 predicted that C3 (NPR avatar) would produce lower eeriness scores on the Ho & MacDorman subscale (D_score) than C2 (photorealistic avatar).

[Insert ANOVA results for B_score across C1, C2, C3. Report F, df, p, η_p^2. Report C2 vs C3 pairwise comparison with Bonferroni-corrected p and effect size.]

H2 — NPR Does Not Reduce Trust

H2 was a null hypothesis: NPR stylisation was not expected to significantly reduce self-reported trust relative to the photorealistic baseline. Motivated by Canales et al. [5], who found equivalent self-reported trust across stylisation levels in a related avatar study, and by Gao et al. [20], who found comparable trust between realistic and stylised virtual agents, H2 predicted that the Competence (E_score), Benevolence (F_score), and Integrity

(G_score) subscales would show no statistically significant difference between C2 and C3.

[Insert ANOVA results for E_score, F_score, G_score. For each: F, df, p, η_p^2. Report C2 vs C3 Bonferroni comparison. If the null holds, interpret absence of significant difference as consistent with H2. If significant, report the direction and discuss.]

H4 — Eeriness Predicts Trust

H4 examined whether eeriness score (B_score) negatively predicts trust scores across all three conditions. A linear regression was specified with B_score as the predictor and each trust subscale composite (E_score, F_score, G_score) as the criterion, entered separately. Condition was included as a covariate, and analyses were conducted at the individual observation level with repeated-measures correction for within-person dependency.

[Insert regression results: β, t, p, R^2 for each trust subscale regressed on B_score. Report whether the eeriness-trust association is consistent across subscales and whether it persists after controlling for condition.]

6.2.4 Primary Behavioural Measure

After completing all three video blocks and questionnaires, participants ranked the three advisors from most to least preferred to consult (P1), without reference to any specific scenario. A rank order was used rather than the strict forced-choice employed by Canales et al. [5] because H3 predicts C1 dominance: if most participants place C1 first in a forced choice, no variance remains to distinguish C2 from C3 in the behavioural measure. A ranking preserves the C2 vs C3 signal regardless of how many participants assign C1 rank 1.

Mean ranks per condition (1 = most preferred, 3 = least preferred) were compared using a Friedman test, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha_{\text{corrected}} = .017$). The theoretically critical comparison is C2 vs C3.

[Insert Friedman result: χ_F^2, df = 2, p. Insert mean rank $\pm$ SD per condition. Insert Wilcoxon results for C1 vs C2, C1 vs C3, and C2 vs C3 with corrected p-values. If C2 vs C3 Wilcoxon is significant, report direction (which condition received lower mean rank, i.e. was preferred more). If not significant, interpret as consistent with H2: NPR stylisation does not reduce implicit preference relative to the photorealistic avatar.]

6.2.5 Secondary Measures

Part H of the questionnaire measured recommendation agreement (H1), perceived warmth and approachability (H2), and perceived influence of visual appearance on advice quality (H2). These items were included as secondary exploratory measures.

[Insert descriptive statistics for Parts H. Note whether recommendation agreement differed across conditions, and whether perceived appearance influence (the meta-cognitive item) differs between

C2 and C3 — participants may report that visual style does not influence their trust judgement even if H2 items reveal implicit differences.]

The propensity-to-trust covariate (Part Q9) was examined as an individual difference moderator. Participants with high baseline propensity to trust may show smaller condition-driven differences; low-propensity participants may be more sensitive to appearance-based cues.

[Insert moderation analysis or descriptive breakdown of trust scores by propensity quartile if sample size permits.]

6.2.6 Pre-Video Baseline Shift

Before each video, participants recorded their preference between the two scenario options (the exploratory pre-video baseline item). Comparing the pre-video preference with the post-video recommendation agreement item (H1) provides a measure of preference shift that isolates the persuasive effect of the advisor's recommendation from the participant's prior inclination. Preference shift was computed per participant per block as the binary change from initial preference to post-video recommendation agreement.

[Insert preference shift rates per condition. If participants are more likely to shift their preference toward the advisor's recommendation in one condition, this suggests a condition-level difference in persuasive credibility that is independent of explicit trust self-report.]

6.3 Discussion

6.3.1 NPR Stylisation and Eeriness

[Interpret H1 result. If confirmed: discuss in terms of uncanny valley hypothesis [3] — NPR abstraction moves the avatar away from the near-human region on the affinity curve. If not confirmed: consider whether the Avaturn baseline was already sufficiently low in eeriness to leave no room for further reduction, or whether the V8 stylisation was not sufficiently distinct.]

6.3.2 Trust Across Conditions

[Interpret H2. The critical question is whether NPR reduces trust relative to photorealistic (C2 vs C3, null expected). Reference Canales et al. [5] if self-reported trust is equivalent between C2 and C3. If trust does differ, consider whether the differential falls on competence, benevolence, or integrity subscales separately, and what this implies for the role of visual style in social judgements of animated speakers. C1 vs C2/C3 differences may be noted briefly as an incidental observation but are not a primary finding.]

6.3.3 Eeriness and Trust as Separable Dimensions

[Interpret H4. If eeriness negatively predicts trust (D_score predicts E/F/G_score), this is consistent with the theoretical account in which uncanny valley discomfort interferes with social trust formation, and supports Alipour et al.'s [18] systematic review finding. Discuss whether the prediction holds equally across competence, benevolence, and integrity subscales or whether it is stronger for one dimension. If H4 is not confirmed, discuss whether this reflects a ceiling effect on the eeriness scale or a genuine independence between eeriness and trust in this context.]

6.3.4 Behavioural and Self-Report Divergence

[Discuss the P1 ranking result in relation to self-report trust. Canales et al. [5] found that the semi-realistic condition attracted the highest forced-choice selection rate despite equivalent self-reported trust. If a similar pattern is observed here — C3 receiving a lower (more preferred) mean rank despite equivalent self-reported trust subscale scores — discuss this dissociation in terms of the implicit-explicit distinction and what it implies for evaluating NPR as a design strategy for avatar-based communication. If C1 receives the lowest mean rank (most preferred) on P1 but self-report trust scores do not separate C1 from C2/C3, note that the behavioural measure is more sensitive to the real-person advantage than the Likert scales.]

6.3.5 Limitations

Several limitations apply to this study. First, the evaluation uses a single NPR condition, V8, so results cannot be generalised to NPR stylisation as a category. A different technique or a different abstraction strength might produce different trust outcomes. Second, the study is conducted online via video stimuli; the absence of spatial presence, interactivity, and stereoscopic depth means that the results reflect trust responses to a speaking head in a video interface, not to an embodied avatar in VR. Third, the Avaturn avatar is a single identity constructed for this study; individual avatar characteristics including perceived age, gender, and attractiveness may interact with the NPR effect in ways the current design cannot decompose. The study recruits participants through [channel], which may oversample populations with higher digital literacy and more prior exposure to AI-generated content, potentially attenuating the uncanny valley response relative to a general population sample.

A further limitation concerns the real-person baseline condition. The C1 stimulus features the thesis researcher as the speaking advisor, meaning participants who knew the researcher personally may have rated C1 with elevated trust scores reflecting social familiarity rather than advisor credibility. Because a recognition check was not included in the questionnaire, it is not possible to identify and separately analyse these participants. C1 ratings should therefore be interpreted as an approximate rather than clean real-person baseline. The theoretically critical comparison between C2 and C3 is unaffected by this confound, given that neither condition features a recognisable person.

The trust subscales use three items per dimension, adapted from the single-item measures of Canales et al. [5] and grounded in the McKnight & Chervany [17] typology. While

this exceeds the minimum for internal consistency assessment, three items per subscale represents a lower bound for robust reliability estimation; a fully validated multi-item instrument would provide a stronger psychometric foundation for the trust outcome measures.

7 Conclusion

7.1 Summary of Contributions

This thesis developed and evaluated a system for applying non-photorealistic rendering to real-time VR avatars across three distinct avatar platforms, with the aim of understanding both the technical feasibility of NPR integration and the perceptual consequences of visual abstraction for avatar trust. The work was structured around three research questions: whether NPR can be integrated into platforms with different rendering architectures, which technique achieves the best trade-off between visual quality and performance, and how visual abstraction affects the perceived trustworthiness of a speaking avatar.

The first contribution is an extensible multi-platform NPR rendering system implemented in Unity 6 with URP, supporting three avatar platforms — Mixamo, Avaturn, and the Meta Avatars SDK — that differ substantially in their shader accessibility and runtime management. Eight stylisation techniques were designed and implemented across this system, each motivated by a specific limitation of its predecessor. The sequence progresses from a geometry-based inverted hull outline through screen-space derivative edge detection, Sobel and Gaussian Sobel colour-boundary detection, hierarchical multi-cue edge fusion, and the isotropic Kuwahara painterly filter, ending in two composite techniques that chain multiple abstraction strategies within a single fragment shader invocation. For the Meta SDK platform, all effects were implemented inside the `AppSpecificPostManipulation` hook of the Extensible Shader Framework, which is the only integration point that does not break the SDK's skinning, LOD management, or PBR material system.

The second contribution is the adaptation of the X-Toon extended toon shading technique [10] to a production avatar context, including a Normal Field Abstraction mechanism on the Mixamo and Avaturn platforms that uses a simplified normal map and TBN-space interpolation to progressively flatten surface shape toward its low-frequency form, and an abstraction compression formula that widens shading bands as the detail axis increases. On the Meta platform, where intermediate lighting terms are not accessible within the post-processing hook, a luminance-based proxy replaces the direct Lambertian dot product, and the light direction is sourced from the URP per-frame global `_MainLightPosition.xyz`, enabling a correct Blinn-Phong specular response.

The third contribution is a runtime tooling infrastructure supporting iterative visual assessment in VR. The fully procedural in-VR parameter panel (`NPREdgeDetectionUI.cs`) exposes over sixty tunable parameters across all technique groups via controller-driven sliders and supports real-time technique switching without shader recompilation. A preset save and recall system allows parameter configurations to be stored to persistent storage and reloaded across sessions, supporting systematic exploration of the parameter space. A material management script (`AvatarShaderSwapper.cs`) resolves the Meta SDK's LOD-driven material reversion problem through a polling coroutine, ensuring that custom NPR assignments persist through all SDK LOD transitions.

7.2 Limitations

Several limitations of the current system are worth noting. The Meta SDK integration is constrained to the `AppSpecificPostManipulation` hook, which fires after the PBR lighting stage; techniques that require access to intermediate lighting terms, per-fragment NdotL, or the tangent frame are therefore not directly realisable on the Meta platform. This limitation required separate implementations for the X-Toon and toon shading techniques, introducing a divergence in the parameter space across platforms that complicates direct visual comparison.

The Kuwahara filter implementation uses the isotropic formulation, which produces rotationally symmetric smoothing kernels. The anisotropic extension of Kyprianidis et al. [13], which aligns the smoothing kernel with local image structure, would produce stronger feature preservation at region boundaries but requires a structure tensor computed in a preceding render pass. Because additional render passes are not feasible within the single-pass Meta SDK hook, and because consistency across platforms was a design goal, the anisotropic formulation was excluded.

Performance on the Meta Quest target hardware was not formally benchmarked in this thesis. The sample counts for the higher-complexity techniques — Gaussian Sobel, the composite Kuwahara pipeline — scale with kernel radius, and the worst-case fragment cost was not profiled across the full range of avatar mesh complexity and viewing distances. This leaves open the question of whether the most sample-intensive techniques run within the Quest's frame budget under the conditions of the user study.

The user study evaluates a single NPR condition (the technique selected through visual assessment) against a photorealistic baseline and a real-person baseline, using a fixed avatar identity and a fixed set of recommendation scenarios. The design does not vary the degree of abstraction within the NPR condition, so it cannot measure how trust responds to the specific strength of stylisation applied. The choice of Avaturn as the study platform means the study measures the effect of NPR applied to a photorealistic AI-reconstructed avatar specifically; responses may differ for more stylised avatar types such as Mixamo.

7.3 Future Work

Several directions extend naturally from this work. The most direct extension is a parametric study of abstraction degree: rather than comparing a single NPR condition to a photorealistic baseline, future work could vary the `_LightingStrength` or `_XToonLightingStrength` blend parameter continuously to measure how trust, uncanny valley ratings, and social presence co-vary with abstraction level. This would produce a more complete mapping of the trust-abstraction curve than the single-point comparison of this thesis.

On the technical side, the Meta SDK hook constraint could potentially be relaxed in future SDK versions that expose earlier pipeline stages; access to per-fragment NdotL and the tangent frame would enable Normal Field Abstraction and direct NdotL toon shading on the Meta platform, eliminating the current cross-platform divergence. Alternatively,

a deferred-injection approach using Unity's Render Graph could apply NPR effects to the Meta avatar in a separate pass outside the SDK hook, at the cost of a second depth sample.

The temporal stability of screen-space edge detection under head motion in VR is a known challenge that was addressed during development through threshold calibration but not through explicit temporal filtering. A moving-average filter on the edge magnitude signal, or a reprojection-based approach that warps edge maps between frames, could reduce the flickering that screen-space derivatives produce at seam boundaries when the avatar is in motion.

Finally, the avatar identity used in the user study was designed to be neutral and gender-unspecified; future work could investigate whether the trust effect of NPR stylisation interacts with avatar identity characteristics such as perceived gender, age, or species, extending the study design toward the broader question of how the form of an avatar mediates social responses to its behaviour.

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Pre-Study Calibration Results

This appendix provides the full results from the pre-study parameter calibration described in Section 4.6. The exercise consisted of five sequential ranking rounds followed by a single final preference question. Figure 1 shows the per-round first-preference distributions, and Figure 2 shows the overall visual appeal votes from the final question. All rendered variants shown to participants appear in Figure 4.18 in the main text.

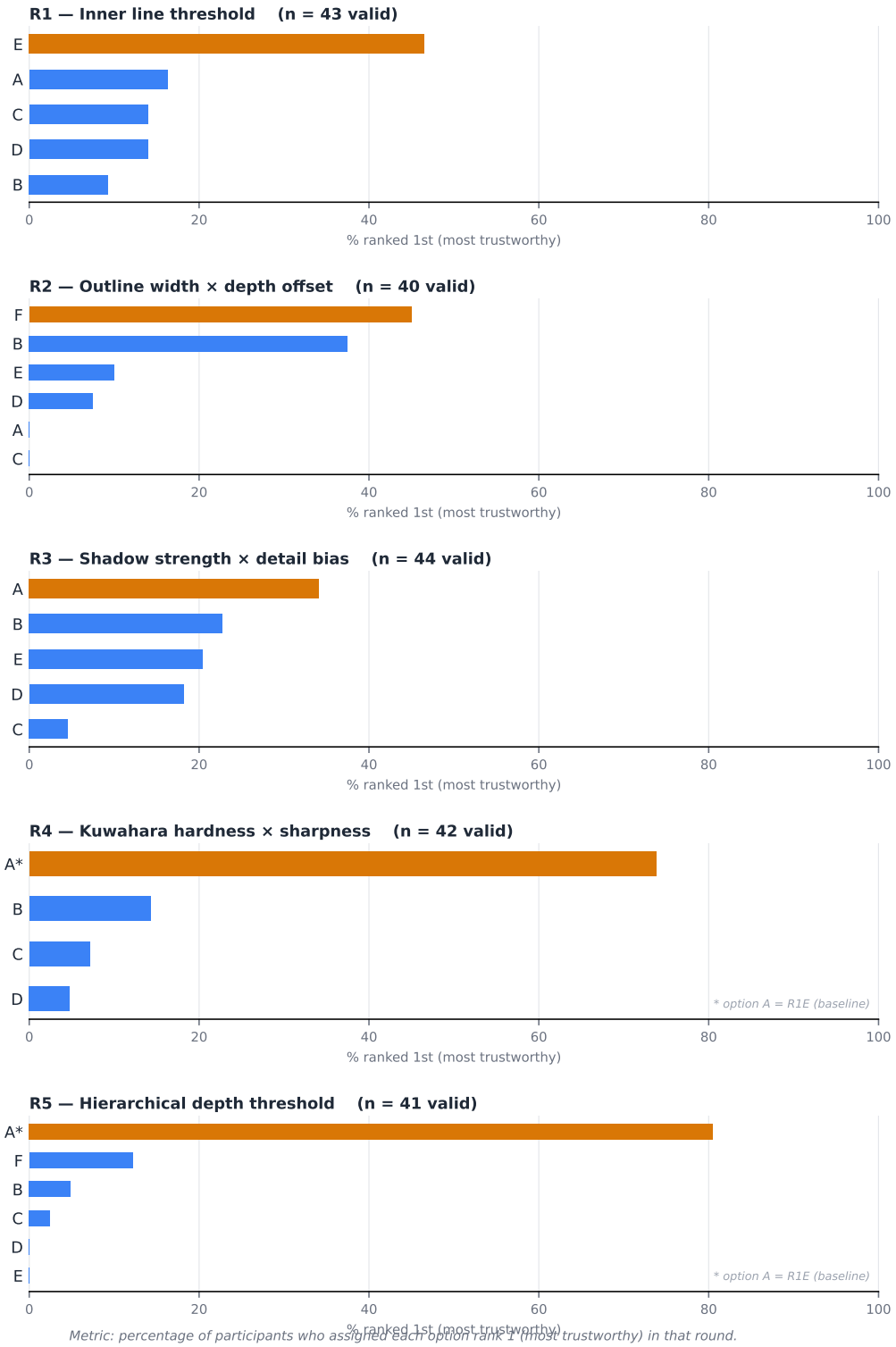

Figure 1: Per-round first-preference distributions from the pre-study ranking exercise. Each bar shows the percentage of valid participants who ranked that configuration first (most trustworthy) within its round. Options are sorted by this percentage, and all panels share the same 0–100 % scale to allow direct visual comparison. The gold bar marks the top-ranked configuration in each round. Options labelled with an asterisk in Rounds 4 and 5 correspond to the carry-forward baseline from Round 1 (R1E, inner line threshold 0.05, no additional post-processing). In both rounds the baseline was preferred by a large majority, and those effects were consequently excluded from the final composite.

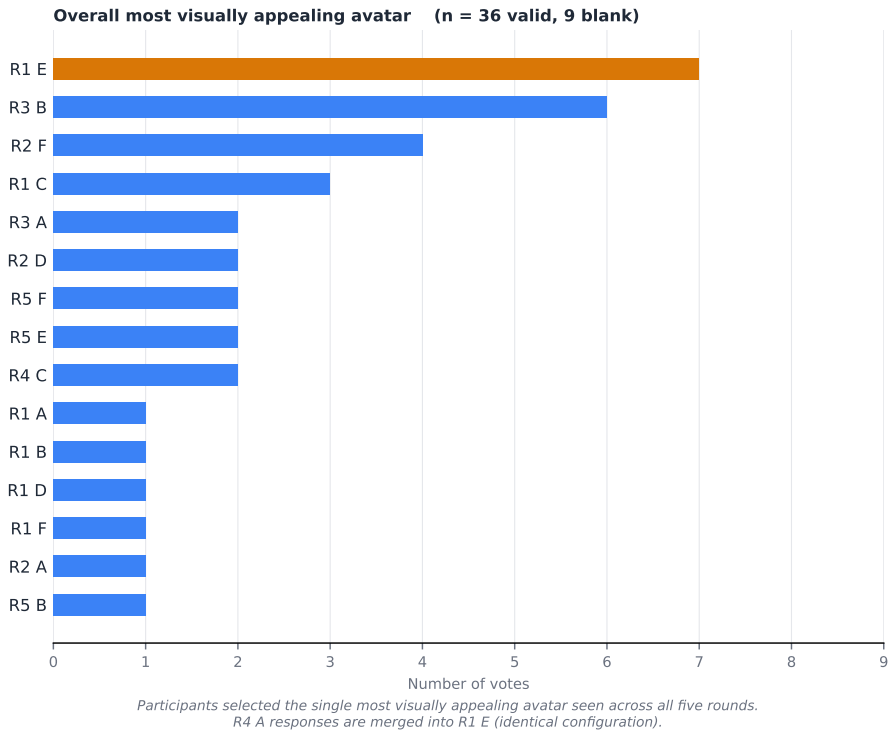

Figure 2: Overall visual appeal votes from the final question of the pre-study. After completing all five rounds, participants were asked to name the single avatar they found most visually appealing across the entire study. Bars show raw vote counts out of 36 valid responses (9 participants left this question blank). One participant selected R4 A, which is identical to R1 E and is merged into that count. R1 E received the most votes overall, confirming the selection made on the basis of the per-round trustworthiness rankings.